Inference, Not Just Optimisation: Streaming Inversion-Free Quasi-Newton Estimation on Dependent Data Streams

Abstract

Streaming quasi-Newton methods fit a model in one pass, as cheaply as stochastic gradient descent but with the accuracy of Newton's method. Their error bars, however, assume that observations are independent, and real streams – traffic counts, air quality, sensor readings – are not. The resulting confidence intervals do not merely lose a little coverage: they converge to a wrong coverage and stay there. On the hourly air-quality stream studied here, a nominal 95% interval covers about 47%.

One change repairs the estimate and the interval together, within the same cost budget: read the stream in contiguous blocks. The block-averaged gradients such a method already computes are exactly the ingredients of a consistent estimator of the variance the intervals need, so the correct statistical object and the computational shortcut coincide – and the valid interval turns out cheaper than the invalid one it replaces. We prove consistency, a rate and a central limit theorem with the correct limit under geometric mixing, derive the exact coverage that dependence-blind intervals converge to, and report the one cost of blocking: a step-size condition it makes non-trivial, with a remedy. Across simulated designs and two real hourly streams, coverage rises from 0.46–0.83 to 0.78–0.95.

1 Introduction

Many estimation problems now arrive as a stream: observations appear one at a time or in small batches, they are too numerous to store, and a decision or a report is required at any moment. Stochastic approximation [78] is the natural tool, and averaged stochastic gradient descent [75, 82] is asymptotically efficient with $O(d)$ work per observation. Its weakness is conditioning: when the curvature of the objective has eigenvalues spread over orders of magnitude, first-order methods crawl [8, 55]. Second-order methods fix the conditioning but appear to cost $O(d^3)$ per step.

A line of work on *inversion-free* stochastic Newton methods removes that obstacle. When the Hessian has the form $\nabla^2 F(\theta) = \mathbb{E}[\alpha(\theta;\xi)\,\Psi(\theta;\xi)\Psi(\theta;\xi)^\top]$—as it does for linear, logistic, softmax, Poisson and ridge regression, and for the geometric median—the running curvature estimate is a sum of rank-one terms, so its inverse can be maintained directly by the Sherman–Morrison rank-one update [84] — also called the Riccati form, the name we use in the proofs — without ever forming or inverting a matrix [6, 9, 12, 33]. Godichon-Baggioni and Werge [34] push this to its natural conclusion: thinning the rank-one updates and taking mini-batches of size d gives an estimator whose *total* cost is $O(dN)$—first-order cost—while remaining asymptotically efficient. A concurrent construction achieves the same budget by masking Hessian columns [36].

All of this theory assumes the stream is independent and identically distributed. The streams that motivate it are not. Hourly traffic volume, pollutant concentrations, queue lengths, and user activity are serially dependent, and after any regression adjustment the *scores* remain serially dependent: in the two real streams we study, the lag-one residual autocorrelation after a fixed weather-and-calendar adjustment is

0.76 (traffic) and 0.92 (air quality). The consequence is not a small correction. Writing $g_i = \nabla_\theta f(\theta^\star; \xi_i)$ for the score at the truth, $\Gamma_k = \mathrm{Cov}(g_0, g_k)$, $H = \nabla^2 F(\theta^\star)$ and

$$S = \sum_{k \in \mathbb{Z}} \Gamma_k \qquad \text{(the \textit{long-run} score covariance)}, \tag{1}$$

any efficient streaming M-estimator satisfies $\sqrt{N}(\hat{\theta}_N - \theta^\star) \rightsquigarrow \mathcal{N}(0, H^{-1}SH^{-1})$, whereas the independent-data theory certifies $\mathcal{N}(0, H^{-1}\Gamma_0 H^{-1})$. The ratio of the two variances, coordinatewise, is a factor κ_j that does not shrink with N. It is not a rounding error: we estimate it at about 2 for the traffic stream and about 20 for the air-quality stream, and at $\kappa_j = 3$ a nominal 95% interval covers 75% of the time *forever*, however long the stream runs.

This paper. We ask what has to change in a streaming inversion-free quasi-Newton method so that both its point estimate and its *uncertainty* are correct on a dependent stream, without leaving the $O(dN)$ budget. Our answer is that one structural choice does both jobs. Consume the stream in *contiguous blocks* of length b and take one preconditioned step per block; place curvature updates on a *gap* of m observations. Then

- the b-averaged gradients the algorithm forms anyway are, up to a factor b, the non-overlapping batch means of the score process, so they *are* an estimator of S—the statistically correct object and the computational shortcut coincide; and
- gapping decorrelates the rank-one curvature terms, which is what makes the inversion-free recursion behave, at matched cost, as it would on independent data.

Neither device substitutes for the other: gapping alone leaves the variance wrong, and blocking alone leaves the curvature recursion aggregating dependent terms. We call the resulting estimator BGSN (blocked–gapped streaming Newton).

Contributions.

1. **Theory for the inversion-free curvature recursion under dependence** (Section 3). We prove almost sure convergence, the rate $\|\theta_T - \theta^\star\|^2 = O(\log N/N)$, and a central limit theorem with the long-run sandwich $H^{-1}SH^{-1}$, for a rank-one Sherman–Morrison curvature recursion with gapped updates on a geometrically mixing stream (Theorems 3 to 6). The independent-data analysis rests on the score being a martingale difference sequence with respect to the algorithm's filtration; dependence destroys that, and existing dependent-data results are first-order, while existing streaming second-order inference assumes conditionally unbiased gradients. A by-product (Proposition 7) is that for this algorithm blocking is *free*: the leading term of the error is $-N^{-1}H^{-1}\sum_{i \le N} g_i$ for *every* b, so the asymptotic variance does not depend on the block length.
2. **A long-run covariance estimator from block gradients, and why it is fast** (Section 4). S is estimated at $O(d^2)$ per block and $O(d^2)$ memory from the same block gradients. The key point is *what* is being blocked: batch means taken over the *iterate* path need blocks long enough for the iterates to mix, which forces $b \asymp N^{3/4}$ and caps the rate at $N^{-1/8}$ [81, 83, 103]; block *gradients* inherit their dependence from the data, so $b \asymp \log N$ suffices and Theorem 10 gives $\widetilde{O}(N^{-1/2})$. This does not contradict the minimax rate $\Theta(N^{-(1-a)/2})$ for Hessian-free covariance estimation from an SGD trajectory with step exponent a [69], which is at best $N^{-1/4}$ over the admissible range $a > 1/2$: a quasi-Newton method is not in that class, because it already maintains curvature and forms block gradients.
3. **An exponentially small block-length bias** (Theorem 10). What makes $b \asymp \log N$ admissible is that the two-scale (flat-top / lugsail) correction $\widehat{S}^{\mathrm{FT}} = \widehat{S}^{\mathrm{BM}} + b(\hat{\Lambda}_1 + \hat{\Lambda}_1^\top)$ has bias $O(\rho^b)$ rather than the

$O(1/b)$ of plain batch means. (A flat-top lag window is one whose weights are flat near lag zero, so its bias picks up only the far tail of the autocovariances rather than shaving every lag.) The correction and this tail-sum property are due to the flat-top literature [72, 73, 85, 90], not to us (Section 4 is explicit about the boundary); what we add is that it is available at *block-gradient* resolution on a dependent stream, and a streaming (dyadic) implementation that needs no advance knowledge of N.

4. **Exactly how badly dependence-blind intervals fail** (Proposition 8). Their asymptotic coverage is $2\Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1$; over a sweep in dependence strength the largest gap between prediction and measurement is 0.0085, and 0.0063 for κ up to 2.9, with no fitted constants (Figure 1). This turns a previously reported qualitative failure into a quantitative one.
5. **A stability condition that blocking creates, and its fix** (Proposition 1). Blocking the gradient makes the step's contraction condition $\gamma_t \|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}} < 2$ non-trivial, where γ_t is the step size, $\bar{H}$ the running curvature estimate used as preconditioner, and $\widehat{H}_t = b^{-1}\sum_{i\in B_t}\alpha_i\Psi_i\Psi_i^\top$ the Hessian of block t alone. On a dependent stream a block of b observations can span far fewer than b distinct covariate directions, so $\widehat{H}_t$ is near-singular and the norm is large. We give the condition, a step-size shift t_0 read off the warm-up window that restores it at $O(c_{\mathrm{w}}d^2)$ one-off cost, and a proof that the shift changes no asymptotic statement. Without it our own method diverges on a regime-switching design; with it that design is stable and its coverage tracks the infeasible oracle interval at every sample size (Tables 2 and 5). We report this because it is a cost of blocking, and it is ours.
6. **Curvature schedules** (Proposition 11). At a matched number of rank-one updates, deterministic gapping incurs excess curvature variance $O(\rho^m)$ whereas the Bernoulli thinning used as the cost knob in the independent-data method incurs $(\kappa_H - 1)/m$. Gapping is thus a strict, if modest, improvement on the existing cost knob, and it makes the per-block cost deterministic.
7. **Evidence** (Section 5). Four dependent designs with exact population targets (three with closed-form S), plus hourly traffic and air-quality streams under two protocols, one of which has an exact ground truth on real covariates.

What we do not claim. The $O(dN)$ budget and the idea of thinning curvature updates are due to Godichon-Baggioni and Werge [34]; blocked or growing mini-batches as a dependence-breaking device are due to Godichon-Baggioni et al. [35]; the long-run sandwich limit under mixing or Markovian sampling is known for first-order stochastic approximation [57, 58, 60, 83]; recursive online estimation of the *instantaneous* asymptotic variance under independent data is due to Godichon-Baggioni [32] and Chen et al. [15], Zhu et al. [103]; online estimation of a *long-run* covariance under dependent sampling is known for first-order methods [81, 83]; online covariance estimation for streaming second-order methods is known under independent data [50, 92]; the two-scale correction is known [73, 85, 90]; the semiparametric efficiency bound under Markovian sampling is due to Li et al. [57], and we attain rather than establish it. Section 6 states each boundary precisely.

2 Setting and notation

We observe a strictly stationary stream $\xi_1, \xi_2, \ldots$ on $\mathbb{R}^p$ and wish to estimate

$$\theta^\star = \arg\min_{\theta\in\mathbb{R}^d} F(\theta), \qquad F(\theta) = \mathbb{E}[f(\theta;\xi)], \tag{2}$$

where $f(\cdot;\xi)$ is a loss. Write $g_i(\theta) = \nabla_\theta f(\theta;\xi_i)$ and $g_i = g_i(\theta^\star)$, so $\mathbb{E}[g_i] = 0$. Let $H = \nabla^2 F(\theta^\star)$, $\Gamma_k = \mathbb{E}[g_0 g_k^\top]$ and let S be as in (1). Throughout, $\Phi(x)$ is the standard normal distribution function, $z_{\alpha/2} = \Phi^{-1}(1-\alpha/2)$, $\|\cdot\|_{\mathrm{op}}$ is the operator norm, and N counts observations while T counts blocks.

Terminology. Five terms recur and are used in their standard senses. The *score* is the gradient of the loss at one observation, $g_i(\theta)$; an *M-estimator* is a minimiser of an empirical average of losses. A *sandwich* covariance is one of the form $H^{-1}\Sigma H^{-1}$; it is *instantaneous* when $\Sigma = \Gamma_0$ and *long-run* when $\Sigma = S$. *Batch means* estimates a variance by averaging within consecutive blocks and taking the sample covariance of the block averages. The strong (or α-) *mixing coefficient* of the stream at lag k is

$$\alpha_{\mathrm{mix}}(k) = \sup\left\{|\mathbb{P}(A\cap B) - \mathbb{P}(A)\mathbb{P}(B)| : A \in \sigma(\xi_i, i \le 0),\ B \in \sigma(\xi_i, i \ge k)\right\}, \tag{3}$$

a measure of how nearly independent the distant past and the distant future are; it is zero for all $k \ge 1$ if and only if the stream is independent.

Nothing below requires the model to be correctly specified. All that is used is that $\theta^\star$ solves $\nabla F(\theta^\star) = 0$ for the *working* loss, so under misspecification $\theta^\star$ is the pseudo-true parameter and $H^{-1}SH^{-1}$ is the corresponding long-run sandwich, in the usual sense of White [96] and Domowitz and White [21]. This matters in practice, because omitted dynamics are one of the commonest ways for a score to become serially correlated (Remark 1).

Remark 1 (Dependence in the covariates is not enough)**.** If the conditional law of y given x is correctly specified and its innovations are independent over time, then (g_i) is a martingale difference sequence and $S = \Gamma_0$ *exactly*, however persistent x is. A genuine long-run-variance problem needs the *score* to be serially correlated: serially correlated errors, a latent dynamic factor, or omitted dynamics. All our designs therefore couple dependent covariates with a dependent error or latent process, and we report the resulting κ_j explicitly. Covariate dependence still matters for the effective sample size of the curvature estimate, which is what Proposition 11 is about.

2.1 The algorithm

Blocks. Partition the stream into contiguous blocks $B_t = \{(t-1)b+1, \ldots, tb\}$, $t = 1, 2, \ldots$, and write $\bar{g}_t(\theta) = b^{-1}\sum_{i\in B_t} g_i(\theta)$ for the block gradient and $\varepsilon_t = \bar{g}_t(\theta^\star)$ for the block score.

Curvature. Let $C = \{i \ge 1 : i \equiv 0 \bmod m\}$ be the *gapped* curvature slots and let $n_t = |C \cap [1, tb]|$ count the rank-one updates made by the end of block t. Maintain an unnormalised accumulator A and a weight W:

$$A_n = A_{n-1} + \iota_n e_{k_n} e_{k_n}^\top + \alpha_{i_n}\Psi_{i_n}\Psi_{i_n}^\top, \qquad W_n = W_{n-1} + 1, \tag{4}$$

where $\alpha_i = \alpha(\theta_{t-1};\xi_i)$ and $\Psi_i = \Psi(\theta_{t-1};\xi_i)$ for $i \in B_t$, e_k cycles through the canonical basis, and $\iota_n = c_\iota\,\varsigma_\star\, n^{-q_{\mathrm{r}}}$ with $q_{\mathrm{r}} \in (0, 1/4)$ is a vanishing ridge that keeps $\lambda_{\min}$ of the curvature estimate bounded below without any matrix operation. Its scale $\varsigma_\star$ is the mean curvature magnitude of the *first* block carrying curvature slots, $\varsigma_\star = (d\,|C\cap B_1|)^{-1}\sum_{i\in C\cap B_1}\alpha_i\|\Psi_i\|^2$, computed once and then frozen. Freezing matters and is not a detail: a scale that tracked the *running* curvature would shrink exactly when the curvature estimate degenerated, and Lemma 2 would then presuppose its own conclusion. With $\varsigma_\star$ fixed, positive and finite almost surely under Assumption 3, the lemma holds unconditionally. The scale exists only to make c_ι dimensionless; setting $\varsigma_\star = 1$ recovers an unscaled ridge and Table 5(d) shows the results are the same. The curvature estimate and preconditioner are

$$\bar{H}_t = A_{n_t}/W_{n_t}, \qquad \bar{H}_t^{-1} = W_{n_t}A_{n_t}^{-1}, \tag{5}$$

and A^{-1} is updated by two rank-one Sherman–Morrison steps per curvature slot, each $O(d^2)$; no matrix is ever inverted. Maintaining A^{-1} rather than $\bar{H}^{-1}$ is what keeps every update exactly rank one despite the changing normalisation W.

Parameter step. With $A_0 = \lambda_0 I$, $W_0 = 1$ (λ_0 a small positive constant in the units of the curvature; Table 5(e) varies it over eight orders of magnitude), and $\bar{H}_{t-1}$ built only from blocks $1, \dots, t-1$, the raw step is

$$\theta_t = \theta_{t-1} - \Pi_t\left[\frac{1}{t}\,\bar{H}_{t-1}^{-1}\bar{g}_t(\theta_{t-1})\right], \qquad \Pi_t[v] = v \cdot \min\left\{1,\ \frac{c_D\,(1+\|\theta_{t-1}\|)}{\|v\|}\right\}, \tag{6}$$

where Π_t is a step-length safeguard, explained next, that is inactive whenever the step is of a reasonable size. The step $1/t$ is what makes the last iterate efficient without averaging; because it is preconditioned by $\bar{H}^{-1}$ there is no learning-rate constant to tune. A weighted averaged variant (step t^{-a}, $a \in (1/2, 1)$, with logarithmically weighted Polyak–Ruppert averaging) is available and reported alongside.

Why a blocked step needs a safeguard. Linearising (6) at $\theta^\star$, ignoring Π_t, gives the error recursion

$$\theta_t - \theta^\star = \left(I - \gamma_t \bar{H}_{t-1}^{-1}\widehat{H}_t\right)(\theta_{t-1} - \theta^\star) - \gamma_t \bar{H}_{t-1}^{-1}\varepsilon_t + o(\|\theta_{t-1} - \theta^\star\|), \tag{7}$$

where $\widehat{H}_t = b^{-1}\sum_{i\in B_t}\alpha_i\Psi_i\Psi_i^\top$ is the Hessian of block t alone. Since $\bar{H}^{-1}\widehat{H}_t$ is a product of two positive semidefinite matrices, its eigenvalues are real and non-negative, so the spectral radius of the linear map in (7) is below one exactly when

$$\gamma_t\,\lambda_{\max}\!\left(\bar{H}_{t-1}^{-1}\widehat{H}_t\right) < 2, \qquad \text{for which} \qquad \gamma_t\,\|\bar{H}_{t-1}^{-1}\widehat{H}_t\|_{\mathrm{op}} < 2 \tag{8}$$

is sufficient, since $\lambda_{\max} \le \|\cdot\|_{\mathrm{op}}$. We work with the operator norm because it is what we can compute and because it is the conservative choice; and we say "spectral radius" rather than "contraction" deliberately, because for a non-symmetric map a spectral radius below one bounds the *asymptotic* growth of powers, not the norm of a single step. What (8) rules out is the regime we actually observe, in which $\gamma_t\lambda_{\max} \gg 2$ and a single step multiplies the error by a large factor. On an *independent* stream with $b \ge d$ the block Hessian has full rank and is close to H, so $\|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}} \approx 1$ and (8) holds already at $\gamma_1 = 1$. On a *dependent* stream it need not: a block of b observations from a persistent stream spans far fewer than b distinct covariate directions, $\widehat{H}_t$ is close to singular, and the norm is inflated by roughly the ratio of the largest to the smallest curvature. We measured $\|H^{-1}\widehat{H}_t\|_{\mathrm{op}}$ at $b = 20$: median 20 on our autoregressive design and 33 on our regime-switching design, with per-block maxima of 55 and 89. So (8) is violated at $\gamma_1 = 1$ on *both*, and the autoregressive designs are lucky rather than safe. Taking a full-length step under those numbers makes the first iterates overshoot by orders of magnitude, and with $\gamma_t = 1/t$ the excursion is undone only at rate $1/t$, so at realistic N it can survive to the end of the stream: without a safeguard our own method's relative variance on the regime-switching design is 1.1×10^{14} times the efficient value, against 1.41 with the safeguard (Table 5(i)), and individual replicates diverge outright. This is not a defect of the base method. It is created by *blocking* the gradient, which is the one thing we add to it, so it is ours to fix and ours to report.

The fix in (6) bounds each step relative to the current iterate, $\|\Pi_t[v]\| \le c_D(1 + \|\theta_{t-1}\|)$ with $c_D = 1$. It is scale-free in θ, costs $O(d)$, and — the point — it is *self-limiting*: it acts only on steps that are actually too long, and leaves every other step exactly as (6) prescribes.

Proposition 1 (The safeguard is eventually inactive)**.** *Let $\mathcal{A} = \{\Pi_t$ is active for only finitely many $t\}$. On $\mathcal{A}$ there is a finite $T_0(\omega)$ with $\theta_t = \theta_{t-1} - \gamma_t\bar{H}_{t-1}^{-1}\bar{g}_t(\theta_{t-1})$ for all $t > T_0$, so the safeguarded and* *unsafeguarded iterates coincide from T_0 on and every asymptotic statement about one holds for the other; in particular Theorems 4 to 6 and Proposition 7 are unaffected. Moreover $\mathbb{P}(\mathcal{A}) = 1$ whenever $\theta_t \to \theta^\star$ and $\|\bar{H}_t^{-1}\|_{\mathrm{op}}\bar{g}_{t+1} \to 0$ almost surely, because then $\|\gamma_t \bar{H}_{t-1}^{-1}\bar{g}_t\| \to 0$ while $c_D(1 + \|\theta_{t-1}\|) \to c_D(1 + \|\theta^\star\|) > 0$.*

We state Proposition 1 conditionally on $\mathcal{A}$ because that is what we can prove, and we say plainly what is therefore *not* proved: we do not establish almost sure convergence for the safeguarded recursion from scratch. The obstruction is real and worth naming — the factor $\min\{1, c_D(1 + \|\theta_{t-1}\|)/\|v\|\}$ depends on the current block's gradient, so it is $\mathcal{F}_t$- rather than $\mathcal{F}_{t-1}$-measurable and correlates with the noise, which breaks the martingale structure the proofs use. A complete treatment is available in the stochastic-approximation literature on expanding truncations [4, 52], at the price of machinery we do not need here. What we do instead is measure $\mathcal{A}$: across every design and every sample size we report, the safeguard binds between 2 and 7 times, and the count does not grow with N (Table 5(i), Table 2), which is the empirical content of $\mathbb{P}(\mathcal{A}) = 1$. A unit test asserts it.

The alternative we rejected, and why. (8) suggests a different remedy: shift the schedule to $\gamma_t = (t+t_0)^{-1}$ with $t_0 = \frac{1}{2}\max_{t\le T_{\mathrm{w}}}\|\bar{H}_{T_{\mathrm{w}}}^{-1}\widehat{H}_t\|_{\mathrm{op}}$ measured on the warm-up blocks, which makes (8) hold at $t = 1$ by construction and has a cleaner proof (it is a finite deterministic reindexing; see Appendix C). We implemented it and it is worse, for a reason worth stating: it slows *every* step, whereas the overshoot needs correcting on only a handful. On the regime-switching design at $N = 200{,}000$ the shift leaves the relative variance at 3.08 against 1.41 for the safeguard, and on a stream short enough that t_0 is comparable to the number of steps — which is the case for our real-data segments — it freezes the estimator altogether. Both variants are in the code and both are reported (Table 5(i)); the safeguard is the default.

Curvature warm-up. The first $\min\{\lceil c_{\mathrm{w}} d/b\rceil, \lfloor \varpi_{\mathrm{w}} N/b\rfloor\}$ blocks are spent on curvature updates only, with no parameter step and with every observation contributing a rank-one update. The first term is the rule; the second caps the warm-up at a fraction $\varpi_{\mathrm{w}} = 0.1$ of the stream and binds only when $c_{\mathrm{w}} d > \varpi_{\mathrm{w}} N$, which is exactly the regime in which calling an $O(d)$-observation warm-up negligible would be false. It matters: on a 4000-observation stream with $d = 11$, spending $c_{\mathrm{w}} d = 2200$ observations on curvature leaves 45 blocked steps and a relative variance of 2.51, whereas capping at 10% leaves 90 steps and a relative variance of 1.28 (Table 5(j)). Our real-data segments are of exactly that length. This consumes $O(d)$ observations and $O(c_{\mathrm{w}} d^3)$ flops—$O(1)$ in N—and changes no asymptotic statement. It is not cosmetic: a $1/t$ step taken while $\bar{H}$ is still uninformative leaves a component that decays like t^{-c} with $c < 1$, and at realistic N that component dominates the sampling error and destroys coverage (Table 5). We regard "do not start the efficient schedule before the curvature estimate is informative" as a practical finding of independent interest.

Cost. Per block after the warm-up: $O(bd)$ for the gradient, $O(d^2)$ for the $\lceil b/m\rceil$ curvature updates and the preconditioned step, and $O(d^2)$ for the covariance accumulators. With $b = m = d$ this is $O(d^2)$ per block of d observations, i.e. $O(d)$ amortised per observation, matching Godichon-Baggioni and Werge [34]. The warm-up adds a one-off $O(c_{\mathrm{w}} d^3)$: it performs $c_{\mathrm{w}} d$ rank-one updates at $O(d^2)$ each, which is the unavoidable cost of producing a $d \times d$ curvature estimate at all and is of the same order as a single matrix factorisation. The stability shift (28) adds a further one-off $O(c_{\mathrm{w}}(d^2 + db)d/b)$, which is 4–5% of the total operation count at the settings of Table 6 and vanishes relative to the streaming pass as N grows. Total time is therefore

$$O\big(dN + c_{\mathrm{w}} d^3\big), \qquad \text{i.e. } O(dN) \text{ once } N \gtrsim c_{\mathrm{w}} d^2, \tag{9}$$

and Table 6 reports the crossover explicitly rather than hiding it in the constant. Memory is $O(d^2)$: the running inverse and two $d \times d$ covariance accumulators. We state both plainly because it is easy to misread an $O(dN)$ *time* claim as an $O(d)$ *memory* claim; it is not, and the same is true of the independent-data method we build on.

3 Theory

3.1 Assumptions

Assumption 1 (Regularity)**.** F is twice continuously differentiable with $\|\nabla^2 F(\theta)\|_{\mathrm{op}} \leq L$ for all θ; $H = \nabla^2 F(\theta^\star) \succ 0$; and $\nabla^2 F$ is Lipschitz on a neighbourhood of $\theta^\star$, so that $\|\nabla F(\theta) - H(\theta - \theta^\star)\| \leq L_\delta \|\theta - \theta^\star\|^2$ for all θ.

Assumption 2 (Expected smoothness and moments)**.** There are $C, C' \geq 0$ with $\mathbb{E}\|g(\theta)\|^2 \leq C + C'(F(\theta) - F(\theta^\star))$ for all θ; there is $\eta > 0$ with $\mathbb{E}\|g(\theta)\|^{2+2\eta} \leq C_\eta (1 + F(\theta) - F(\theta^\star))^{1+\eta}$; and $\theta \mapsto \mathbb{E}[g(\theta) g(\theta)^\top]$ is continuous at $\theta^\star$.

Assumption 3 (Curvature form)**.** $\nabla^2 F(\theta) = \mathbb{E}[\alpha(\theta;\xi)\Psi(\theta;\xi)\Psi(\theta;\xi)^\top]$ with $\alpha \geq 0$; $\mathbb{E}\|\alpha(\theta;\xi)\Psi\Psi^\top\|_{\mathrm{op}}^{\eta'} \leq C_{\eta'}$ for some $\eta' > 2$ uniformly on a neighbourhood of $\theta^\star$; and $\theta \mapsto \alpha(\theta;\xi)\Psi(\theta;\xi)\Psi(\theta;\xi)^\top$ is Lipschitz in θ in L^2 near $\theta^\star$.

Assumption 4 (Identifiability)**.** $\theta^\star$ is the only stationary point of F outside which the gradient points away from it: for every $\eta > 0$, $\inf_{\|\theta - \theta^\star\| \geq \eta} \langle \nabla F(\theta), \theta - \theta^\star \rangle > 0$.

Assumption 4 holds whenever F is convex with a unique minimiser, which covers every objective we consider (least squares, logistic, ridge, softmax, the geometric median). It is what turns "the gradient step cannot keep moving" into "$\theta_t \to \theta^\star$", and the independent-data analysis needs it too.

Assumption 5 (Geometric mixing)**.** $(\xi_i)_{i \geq 1}$ is strictly stationary and its mixing coefficient (3) decays geometrically: $\alpha_{\mathrm{mix}}(k) \leq K c_{\mathrm{mix}}^k$ for some $c_{\mathrm{mix}} \in (0,1)$, $K < \infty$. Moreover $\mathbb{E}\|g_1\|^{4+\nu'} < \infty$ for some $\nu' > 0$, $\mathbb{E}\|g_1\|^{2+\nu} < \infty$ for some $\nu > 0$ (without loss of generality $\nu \leq 2\eta$), and $S \succ 0$. The fourth moment with a margin is used only for the variance of the covariance estimator in Theorem 10(iii): the products $b\, \bar{g}_{tj} \bar{g}_{tk}$ whose autocovariances that bound needs are quadratic in the scores, and Davydov's inequality applied to them requires $2 + \delta$ moments of the products, i.e. $4 + 2\delta$ of the scores. Plain fourth moments would leave the argument one ϵ of integrability short, which is why we ask for the margin rather than quietly using it.

Assumptions 1 to 3 are those of the independent-data analysis [34], with Assumption 3 strengthened from a moment bound to a local Lipschitz condition that we need because the curvature terms are now dependent. Assumption 5 is the only genuinely new assumption; a geometrically ergodic Markov chain [63] satisfies it, as do stationary causal ARMA/GARCH-type recursions with geometrically decaying dependence in Wu's sense [99, 101]. Davydov's covariance inequality [18, 76] converts Assumption 5 into a geometric bound on the score autocovariances that we use repeatedly:

$$\|\Gamma_k\|_{\mathrm{op}} \;\leq\; C_\Gamma\, \rho^{|k|}, \qquad \rho = c_{\mathrm{mix}}^{\nu/(2+\nu)} \in (0,1). \tag{10}$$

In particular S in (1) converges absolutely.

3.2 The curvature recursion

The first result is where dependence bites. Under independence the terms accumulated in (4) form (conditionally on the iterates) an independent array and standard strong laws apply. Under Assumption 5 they are dependent, and the gapped design is what restores control: the subsequence indexed by C is itself stationary with mixing coefficients $\alpha_{\mathrm{mix}}(jm) \le Kc_{\mathrm{mix}}^{jm}$.

Lemma 2 (The curvature estimate is never ill-conditioned)**.** *Under Assumption 3, with $\iota_n = c_\iota \varsigma_\star n^{-q_{\mathrm{r}}}$, $\varsigma_\star > 0$ fixed and $q_{\mathrm{r}} \in (0,1)$, there is a finite random $c_1 > 0$ with*

$$\lambda_{\min}(\bar{H}_t) \;\ge\; c_1\, n_t^{-q_{\mathrm{r}}} \qquad \textit{and hence} \qquad \|\bar{H}_t^{-1}\|_{\mathrm{op}} \;\le\; c_1^{-1} n_t^{q_{\mathrm{r}}} = O(t^{q_{\mathrm{r}}}) \qquad \textit{for all } t,\ \textit{a.s.}$$

This is the whole purpose of the vanishing ridge, and it needs no assumption on the iterates: it makes $\sum_t \gamma_t^2 \|\bar{H}_{t-1}^{-1}\|_{\mathrm{op}}^2 = \sum_t O(t^{2q_{\mathrm{r}}-2}) < \infty$, the square-summability half of the Robbins–Monro condition, for *every* t rather than eventually, and without any eigenvalue computation.

Theorem 3 (Curvature rate)**.** *Under Assumptions 1, 3 and 5, suppose $\theta_t \to \theta^\star$ almost surely with $\|\theta_t - \theta^\star\| = O(t^{-1/2}\log^c t)$ a.s. for some $c < \infty$. Then for every $\nu_H < q_{\mathrm{r}}$,*

$$\|\bar{H}_t - H\|_{\mathrm{op}} = O(t^{-\nu_H}) \quad \textit{and} \quad \|\bar{H}_t^{-1} - H^{-1}\|_{\mathrm{op}} = O(t^{-\nu_H}) \qquad \textit{a.s.}$$

Remark 2 (Order of the proofs)**.** Theorem 3 presupposes a rate for the iterates while the results below use Theorem 3, so the logical order must be stated. It is a four-rung ladder, each rung using only the ones below it: (i) Lemma 2, from the ridge alone, with no assumption on the iterates; (ii) Theorem 4, $\theta_t \to \theta^\star$ a.s., using only Lemma 2; (iii) a preliminary rate $\|\theta_t - \theta^\star\| = O(t^{-1/2}\log^{1+\varpi} t)$ a.s. (Lemma 16, proved in Appendix A.4), from (24) with $\bar{H}_{t-1}^{-1}$ bounded by Lemma 2 and $\bar{H}_t \to H$ — consistency only, which follows from (ii); (iv) Theorem 3, and then Theorems 5, 6, 9 and 10. Appendix A is organised in this order, and rung (iii) is where the restriction $q_{\mathrm{r}} < 1/4$ is used — everything else needs only $q_{\mathrm{r}} < 1/2$.

3.3 Consistency, rate, and the central limit theorem

The independent-data proofs use that ε_t is a martingale difference with respect to $\mathcal{F}_{t-1} = \sigma(\xi_1, \ldots, \xi_{(t-1)b})$. Under Assumption 5 it is not: $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] \ne 0$, and no choice of block length repairs this, because the conditional bias of a block average is $O(1/b)$ and does not vanish at fixed b.

Our substitute is Gordin's martingale–coboundary decomposition. Under Assumption 5 the conditional expectations $\mathbb{E}[\varepsilon_n \mid \mathcal{F}_0]$ decay geometrically in L^2, so

$$\varepsilon_t = m_{t+1} + z_t - z_{t+1}, \qquad \mathbb{E}[m_{t+1} \mid \mathcal{F}_t] = 0, \quad \mathbb{E}\|z_1\|^2 < \infty, \tag{11}$$

with (m_t) stationary and square integrable and $\mathbb{E}[m_1 m_1^\top] = S/b$ (Lemma 13). Every weighted sum $\sum_t \gamma_t \bar{H}_{t-1}^{-1}\varepsilon_t$ then splits into a square-integrable martingale, which converges almost surely because $\sum_t \gamma_t^2 \|\bar{H}_{t-1}^{-1}\|_{\mathrm{op}}^2 = \sum_t O(t^{2q_{\mathrm{r}}-2}) < \infty$ by Lemma 2, and a telescoping coboundary, which converges because one rank-one Riccati update moves the preconditioner by only $O(1/t)$ (Lemma 15). This is what makes the argument work with weights that are *random and adapted* — unavoidable here, since the weights contain $\bar{H}_{t-1}^{-1}$ — and it is why the long-run covariance S, rather than the instantaneous Γ_0, is the object that appears: it is the second moment of the martingale part in (11). Appendix A gives the details.

Theorem 4 (Almost sure convergence)**.** *Under Assumptions 1 to 5, the iterates (6) satisfy $\theta_t \to \theta^\star$ almost surely, for every fixed block length $b \ge 1$ and gap $m \ge 1$.*

Theorem 5 (Rate)**.** *Under the assumptions of Theorem 4,* $\|\theta_t - \theta^\star\|^2 = O(\log(N_t)/N_t)$ *almost surely, where* $N_t = bt$*.*

Theorem 6 (Central limit theorem with the long-run sandwich)**.** *Under Assumptions 1 to 5, for each* fixed *block length* $b \geq 1$*,*

$$\sqrt{N_T}\,(\theta_T - \theta^\star) \rightsquigarrow \mathcal{N}\Big(0,\ H^{-1}SH^{-1}\Big), \qquad N_T = bT,$$

with S as in (1). The same limit holds for the weighted averaged variant. If instead $b = b_N \to \infty$ with $b_N = o(\sqrt{N})$, the same limit holds provided the random constants Λ of Lemma 14 are bounded in probability uniformly in N along the sequence b_N.

We separate the two cases because only the first is proved unconditionally, and the difference matters: Theorem 10 takes $b \asymp \log N$, so the interval we actually recommend lives in the second case. The proof of the fixed-b statement is almost sure and goes through Lemma 14, whose conclusion carries a finite random constant; for a triangular array in which b changes with N, "finite for each N" is not enough, and the naive route around it — Cauchy–Schwarz on $\mathbb{E}\|H^{-1} - P_{t-1}\|_{\mathrm{op}}\|\varepsilon_t\|$ — is exactly the bound that the proof of term (A) in Appendix A shows is too weak, by a factor $T^{1/2}$. Establishing the uniformity would require redoing Lemma 14 with explicit constants in b, which we have not done. What we can say is that the condition is about the remainder terms and not about the limit, that growing b makes the block scores *less* dependent rather than more, and that empirically coverage is flat in b over $b \in [10, 320]$ at $N = 200{,}000$ (Table 5(b)), which is the composition the two theorems need. Section 7 lists this as an open point rather than burying it.

The condition $b = o(\sqrt{N})$ is what makes the number of steps $T = N/b$ large enough for the $1/t$ schedule to erase the initial condition: with $\gamma_t = 1/t$ the effect of θ_0 decays like $1/T = b/N$, so its contribution to $\sqrt{N}(\theta_T - \theta^\star)$ is of order $b/\sqrt{N}$. It is amply satisfied by the $b \asymp \log N$ of Theorem 10, and it is the reason Table 5(b) shows the point estimate degrading once b becomes a substantial fraction of $\sqrt{N}$.

Proposition 7 (Blocking is free)**.** *Under the assumptions of Theorem 6,*

$$\theta_T - \theta^\star = -\frac{1}{N_T}H^{-1}\sum_{i \leq N_T} g_i + R_T, \qquad \|R_T\| = o_p(N_T^{-1/2}),$$

so the asymptotic variance in Theorem 6 is the same for every block length b.

Proposition 7 matters for the design argument. It says that, for this algorithm, averaging the gradient over a block costs nothing statistically—the leading term is the same sample average of scores whatever b is—which is why we are free to choose b for the covariance estimator. It also answers the objection that single-sample streaming updates are more sample-efficient than batched ones [58]: that is true for a step size $\gamma_t \asymp t^{-a}$ with $a < 1$, and false here.

3.4 The variance-inflation factor and the exact failure of dependence-blind intervals

Proposition 8 (Coverage of dependence-blind intervals)**.** *Let* $\kappa_j = \big[H^{-1}SH^{-1}\big]_{jj} \,/\, \big[H^{-1}\Gamma_0 H^{-1}\big]_{jj}$ *and let* $\widehat{\Gamma}_0 \to_p \Gamma_0$*. Then the interval* $\theta_{T,j} \pm z_{\alpha/2}\{[\bar{H}^{-1}\widehat{\Gamma}_0\bar{H}^{-1}]_{jj}/N\}^{1/2}$ *has asymptotic coverage*

$$2\,\Phi\big(z_{\alpha/2}/\sqrt{\kappa_j}\big) - 1.$$

The proof is one line, and the content is entirely in κ_j: dependence-blind intervals do not merely lose a little coverage, their coverage converges to a computable number below the nominal level. For the homogeneous autoregressive design of Section 5—covariates $x_t = \phi x_{t-1} + \cdots$, errors $u_t = \psi u_{t-1} + \cdots$, $x \perp u$—one gets $\Gamma_k = \sigma_u^2 (\phi\psi)^{|k|} \Sigma_x$ and $H = \Sigma_x$, hence

$$\kappa_j = \frac{1 + \phi\psi}{1 - \phi\psi} \quad \text{for every } j, \tag{12}$$

so at $\phi = \psi = 0.7$ a nominal 95% interval covers 74.9%. (12) is textbook geometric summation and we present it only because it makes Proposition 8 checkable with no free parameters (Figure 1); the qualitative failure has been observed before [60], whose simulated coverage "hovers around 85%" at $\kappa = 2$, against our prediction of 83.4%.

4 Estimating the long-run covariance from block gradients

The algorithm forms $\bar{g}_t(\theta_{t-1})$ once per block and then discards it. Retain it for two $O(d^2)$ accumulations. Over a retained window $\mathcal{T}$ of T' blocks (see below), define

$$\widehat{S}^{\mathrm{BM}} = \frac{b}{T'} \sum_{t \in \mathcal{T}} \bar{g}_t \bar{g}_t^\top, \qquad \hat{\Lambda}_1 = \frac{1}{T' - 1} \sum_{t,\, t+1 \in \mathcal{T}} \bar{g}_t \bar{g}_{t+1}^\top, \qquad \widehat{S}^{\mathrm{FT}} = \widehat{S}^{\mathrm{BM}} + b\big(\hat{\Lambda}_1 + \hat{\Lambda}_1^\top\big). \tag{13}$$

$\widehat{S}^{\mathrm{BM}}$ is the non-overlapping batch-means estimator at block resolution. $\widehat{S}^{\mathrm{FT}}$ is the two-scale (flat-top, or "lugsail") correction. It has the same expectation as $2\widehat{S}^{\mathrm{BM}}(2b) - \widehat{S}^{\mathrm{BM}}(b)$ — Richardson extrapolation in $1/b$ — and coincides with it exactly if $\hat{\Lambda}_1$ is formed from disjoint rather than all adjacent block pairs; we use all adjacent pairs because it has the same expectation and a smaller variance. Either way the expectation is that of a lag-window estimator with weights 1 for $|k| \le b$ and $(2b - |k|)/b$ for $b < |k| < 2b$. **This device is not new**: it is the flat-top window of Politis and Romano [73], refined by Politis [72] and recast as "lugsail" by Vats and Flegal [90], and Singh et al. [85, Eqs. (15) and (17)] apply exactly these two forms to stochastic gradient descent under independent data. Nor is the $O(\rho^b)$ bias order a new mechanism: for a stationary sequence whose autocovariances decay geometrically it is the classical tail-sum property of a flat-top window. What we contribute is that it is available *at block-gradient resolution on a dependent stream*, where it is what makes $b \asymp \log N$ admissible and therefore what buys the rate, together with the streaming implementation below.

Confidence intervals. With $u_j = \bar{H}^{-1} e_j$ (one $O(d^2)$ product using the already-maintained inverse), report $\theta_{T,j} \pm z_{\alpha/2} \{u_j^\top \widehat{S}^{\mathrm{FT}} u_j / N\}^{1/2}$. Only the requested coordinates cost anything; the full $d \times d$ sandwich is never needed.

Retained window (a streaming detail that matters). Early blocks have θ_{t-1} far from $\theta^\star$ and inflate (13). Discarding a fixed fraction requires knowing N in advance, which a streaming method does not. We instead keep two accumulators covering the current and previous dyadic windows $[2^k, 2^{k+1})$ and report from their union, so at any time at least half and at most three quarters of the blocks are retained and the discarded set is a prefix. This answers, in our setting, the open problem of an online implementation of equal-batch-size estimators raised by Singh et al. [85, Remark 4].

Theorem 9 (L^2 rate)**.** *Under the assumptions of Theorem 4,* $\mathbb{E}\|\theta_t - \theta^\star\|^2 = O(\log(N_t)/N_t)$*, uniformly over* $b \le C \log N$.

The L^2 statement is separate from the almost sure one because the drift analysis in Theorem 10 needs a moment bound and an almost sure rate with an unspecified random constant does not supply one. All remainder bounds in Appendix A are uniform over $b \le C \log N$, which is the only regime in which we use them.

Theorem 10 (Long-run covariance estimation)**.** *Let Assumptions 1 to 5 hold and let $b = b_N \to \infty$ with $b_N \le C \log N$. Write $\Delta = \sum_{k\ge1} k\,(\Gamma_k + \Gamma_k^\top)$. Then, entrywise,*

(i) $\mathbb{E}\big[\widehat{S}^{\mathrm{BM}}\big] = S - \Delta/b + O(\rho^b) + O(\tau_{b,N})$*;*

(ii) $\mathbb{E}\big[\widehat{S}^{\mathrm{FT}}\big] = S + O(\rho^b) + O(\tau_{b,N})$*;*

(iii) $\operatorname{Var}\big[\widehat{S}^{\mathrm{FT}}_{jk}\big] = O(b/N)$*; and consequently*

(iv) $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} = O_p\big(d\sqrt{b/N} + \rho^b + \tau_{b,N}\big)$*, which at fixed d and $b \asymp \log N$ gives* $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} = \widetilde{O}_p(N^{-1/2})$*, whereas* $\|\widehat{S}^{\mathrm{BM}} - S\|_{\mathrm{op}} = O_p(d\sqrt{b/N} + 1/b)$ *is minimised at $b \asymp (N/d^2)^{1/3}$ and gives $O_p((d^2/N)^{1/3})$.*

The factor d in (iv) is the cost of converting the entrywise bound (iii) into an operator norm; we make no attempt to remove it, and every rate statement in this paper is at fixed d. Here $\tau_{b,N} = (b+d)\log^2 N/N$ is the contribution of the drift $\theta_{t-1} \ne \theta^\star$: the b comes from the parameter error and the d from the fact that the block curvature $\widehat{H}_t$ is a b-sample estimate of a $d \times d$ matrix, so that $\mathbb{E}\|\widehat{H}_t\delta\|_{\mathrm{op}}^2 \lesssim (1 + d/b)\|H\delta\|_{\mathrm{op}}^2$. Both are dominated by the $\sqrt{b/N}$ standard deviation whenever $b + d = o(\sqrt{N})$ up to logarithms. Consequently $\sqrt{N}\,(\theta_{T,j} - \theta_j^\star)/\{u_j^\top \widehat{S}^{\mathrm{FT}} u_j\}^{1/2} \rightsquigarrow \mathcal{N}(0,1)$ and the intervals above have asymptotically exact coverage.

Remark 3 (Why blocking gradients beats blocking iterates)**.** Batch-means estimators for stochastic approximation are usually built from the *iterate* path [15, 81, 83, 85, 103]. Under dependent sampling this forces long blocks: the iterates θ_k move by $\gamma_k \asymp k^{-a}$, so a block must span $\sum_{k=t+1}^{t+b} \gamma_k$ of parameter movement before its average behaves like an independent draw, which requires $b \asymp N^a$ and yields rates of order $N^{-1/8}$ [e.g. 83, Prop. 2 and Cor. 2]. Block *gradients* do not have this problem: their serial dependence is inherited from the data process, whose mixing time is a constant, and the iterate's own movement enters only through the $O(b/N)$ drift term in Theorem 10. This is the substantive reason for the rate difference, and it is available only to an algorithm that forms block gradients and keeps a curvature estimate. It is also why the minimax rate $\Theta(N^{-(1-a)/2})$ for *Hessian-free* covariance estimation from an SGD trajectory [69] does not apply: a quasi-Newton method is outside that class.

Remark 4 (Positive semidefiniteness)**.** $\widehat{S}^{\mathrm{FT}}$ is a difference of positive semidefinite matrices and need not be positive semidefinite; the same caveat attaches to all flat-top and lugsail estimators [85, Remark 5]. For marginal intervals only the scalars $u_j^\top \widehat{S}^{\mathrm{FT}} u_j$ must be positive, and we fall back to $u_j^\top \widehat{S}^{\mathrm{BM}} u_j$ if one is not. We report the frequency of that fallback in every experiment; it is 0.0% throughout.

Proposition 11 (Gapping versus Bernoulli thinning at matched cost)**.** *Let $Y_i = \alpha(\theta^\star;\xi_i)\Psi_i\Psi_i^\top$, $\Gamma_k^H = \operatorname{Cov}(Y_0, Y_k)$, $S^H = \sum_k \Gamma_k^H$ and $\kappa_H = \|S^H\|_{\mathrm{op}}/\|\Gamma_0^H\|_{\mathrm{op}}$. From a stream of length N, take n rank-one curvature updates either at a deterministic gap $m = N/n$ or by Bernoulli($p = n/N$) thinning of every index. Then, as $n \to \infty$,*

$$n \operatorname{Var}\big[\widehat{H}^{\mathrm{gap}}\big] = \Gamma_0^H + O(\rho^m), \qquad n \operatorname{Var}\big[\widehat{H}^{\mathrm{Bern}}\big] = \Gamma_0^H + \tfrac{1}{m}\big(S^H - \Gamma_0^H\big).$$

So the excess over the independent-data benchmark Γ_0^H/n is exponentially small in m for gapping and of order $(\kappa_H - 1)/m$ for Bernoulli thinning. Gapping also makes the per-block work deterministic. We verify both statements in Figure 4. We are explicit that this is a curvature-side improvement: $\bar{H}$ enters the limit distribution only through the rate ν_H of Theorem 3, so the end-to-end effect on $\hat{\theta}$ is second order, and we do not claim otherwise.

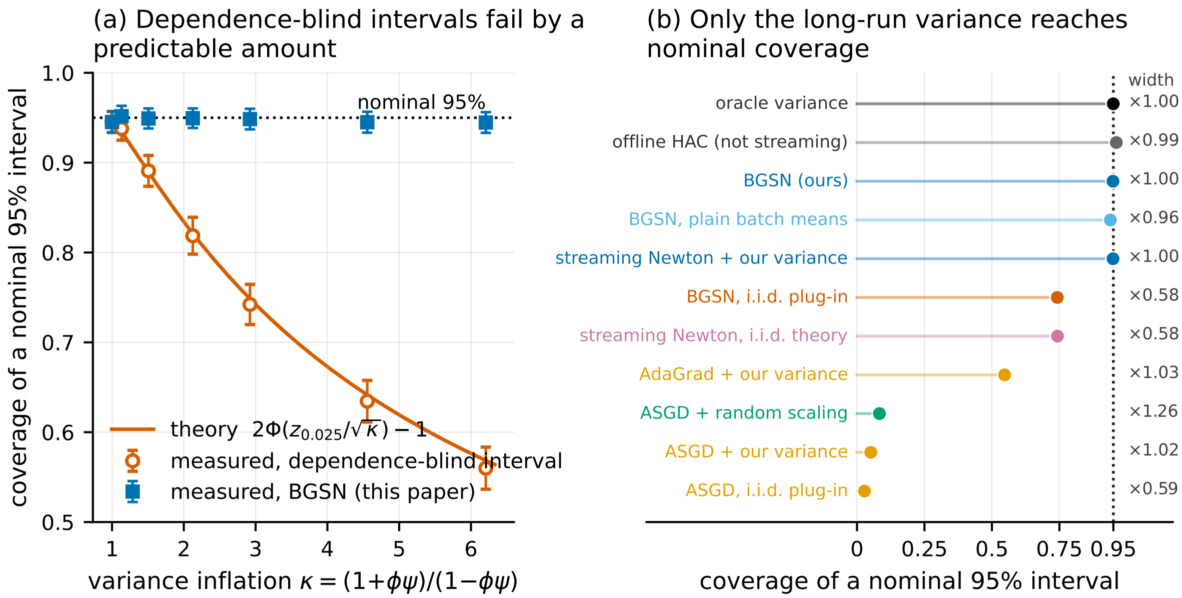

Figure 1: Dependence-blind confidence intervals fail by a computable amount, and only estimating the long-run score covariance repairs them. (a) Coverage of nominal 95% intervals on the homogeneous autoregressive design as the dependence strength varies, against the parameter-free prediction of Proposition 8; error bars are ± 1.96 Monte Carlo standard errors. (b) Coverage of every method on that design at $\phi = \psi = 0.7$ ($\kappa = 2.92$), one row per method; the dotted line is the nominal level and the number at the right is the interval width relative to the infeasible oracle interval. Read it in three groups. The five rows at the top all account for the autocovariances of the score — the infeasible oracle, the offline reference, and the three streaming estimators that use our long-run covariance — and they are the rows that reach nominal, at width ≈ 1. The next two use the instantaneous covariance the independent-data theory prescribes; they are narrower by the factor $\sqrt{\kappa} = 1.71$ and they miss, by the amount Proposition 8 predicts. The bottom four are first-order methods whose *point* estimates have not converged on this ill-conditioned design (their $\sqrt{N}$ error is 1.03 to 9.77 times the efficient value), so their coverage is dominated by that and not by which covariance they use; Table 3 is where they become informative about inference.

5 Experiments

Every number quoted below is generated from the saved results by `experiments/make_tables.py` and injected as a macro, so the prose cannot drift from the data. Monte Carlo standard errors are shown or stated everywhere.

Designs. Four simulated dependent designs with $d = 20$, described in full in Appendix B.1: **AR-hom** (autoregressive covariates and autoregressive errors, closed-form S, the same variance inflation $\kappa = 2.92$ in every coordinate); **AR-het** (coordinatewise-varying persistence, closed-form S, κ_j ranging over $[1.00, 4.41]$, so a scalar correction cannot pass); **Markov** (covariates are a Markov-modulated Gaussian mixture with $d+1$ regimes, hence non-Gaussian with discrete latent state and regime dependence, closed-form S, $\kappa_j \equiv 7.55$); **Logistic** (a correctly specified marginal logistic model with a serially dependent latent error, so $\theta^\star$ is exactly the generating value while the score is autocorrelated; S from an independent long simulation, relative Monte Carlo error 7.5×10^{-3}). All covariate covariances are the ill-conditioned design of the base paper (cond(H) up to 400); Table 3 adds a well-conditioned counterpart, and two real hourly streams are described under Table 4.

Table 1: Coverage of nominal 95% confidence intervals, interval width, and estimation error on four dependent designs (d = 20, N = 200,000, block length b = 20, R = 300 replications; R = 60 for the offline reference). *cov*: coverage, pooled over the d coordinates (Monte Carlo standard error $\leq$ 0.008 throughout). *w*: interval width divided by the width of the infeasible oracle interval. *err*: $\sqrt{N}$ RMSE divided by the asymptotic standard deviation, averaged over coordinates; 1.00 is efficient. κ is the variance inflation caused by dependence. Only methods that estimate the *long-run* score covariance reach nominal coverage; methods using the instantaneous covariance that the independent-data theory prescribes undercover by an amount predicted exactly by Proposition 8. **Read the first-order rows with care**: these designs are ill-conditioned (that is what streaming second-order methods are for), so those methods' *point* estimates have not converged — their *err* column is several times 1 — and their coverage is dominated by that, not by which covariance they use. Table 3 repeats the AR-hom column on a well-conditioned design where every point estimator is efficient, which is where the first-order rows become informative about inference. The Markov column is the one design where *no* streaming method's point estimate has converged at this N; Table 2 resolves it by sweeping N against the oracle interval. ASGD is averaged stochastic gradient descent (Polyak–Ruppert averaging of the iterates); AdaGrad is the diagonally rescaled first-order method; *random scaling* is the variance-estimator-free studentisation of Lee et al. [54].

	AR-hom			AR-het			Markov			Logistic		
method	cov	w	err	cov	w	err	cov	w	err	cov	w	err
dependence	$\kappa \in [2.92, 2.92]$			$\kappa \in [1.00, 4.41]$			$\kappa \in [7.55, 7.55]$			$\kappa \in [2.27, 2.32]$		
BGSN (this paper)	**0.949**	1.00	1.01	**0.950**	1.00	1.01	0.890	1.02	1.27	0.783	0.84	1.33
with plain batch means	**0.939**	0.96	1.01	**0.944**	0.98	1.01	0.856	0.92	1.27	0.768	0.81	1.33
with the i.i.d. plug-in variance	0.742	0.58	1.01	0.832	0.73	1.01	0.460	0.38	1.27	0.581	0.55	1.33
streaming Newton, i.i.d. theory	0.743	0.58	1.01	0.834	0.73	1.01	0.460	0.38	1.27	0.584	0.55	1.33
with our long-run variance	**0.949**	1.00	1.01	**0.951**	1.00	1.01	0.890	1.02	1.27	0.781	0.84	1.33
ASGD + our long-run variance	0.051	1.02	9.77	0.271	1.06	50.34	0.239	1.09	7.95	0.714	1.01	1.91
ASGD, i.i.d. plug-in	0.028	0.59	9.77	0.195	1.03	50.34	0.073	0.38	7.95	0.520	0.66	1.91
ASGD + random scaling	0.083	1.26	9.77	0.679	8.51	50.34	0.230	1.73	7.95	0.925	2.16	1.91
AdaGrad + our long-run variance	0.547	1.03	2.99	**0.947**	1.01	1.03	0.533	1.00	2.76	0.911	1.01	1.15
offline HAC *(not streaming)*	**0.961**	0.99	0.95	**0.955**	0.99	0.98	0.968	0.98	0.85	**0.945**	1.00	1.03
oracle variance *(infeasible)*	**0.950**	1.00	1.01	**0.949**	1.00	1.01	0.886	1.00	1.27	0.858	1.00	1.33

Methods compared. Our estimator BGSN; the same point estimator with plain batch means or with the i.i.d. plug-in variance (so that the interval, and only the interval, changes); the base independent-data streaming Newton method with its own curvature schedule (m = 1, Bernoulli $p = 1/d$) and its own prescribed plug-in variance; averaged SGD and streaming AdaGrad with tuned step sizes, each with the i.i.d. plug-in, with our long-run variance, or with random scaling (a studentisation that needs no variance estimate at all: it divides by a functional of the iterate path whose limit distribution is known and tabulated, 54); an offline HAC reference (heteroskedasticity- and autocorrelation-consistent: the textbook two-pass sandwich with a Bartlett-kernel long-run covariance, which stores the whole stream and is therefore not a streaming method); and an infeasible oracle-variance interval that uses the true H and S. First-order baselines are given the *true* Hessian for their sandwich — an advantage no practitioner has — and their step-size constants are tuned by grid search on pilot streams disjoint from the evaluation streams (Appendix B.3). Our method has no learning rate to tune: its step size is fixed at $1/t$ by the preconditioner, with the step-length safeguard of (6) the only other free quantity and its constant fixed at $c_D = 1$ throughout.

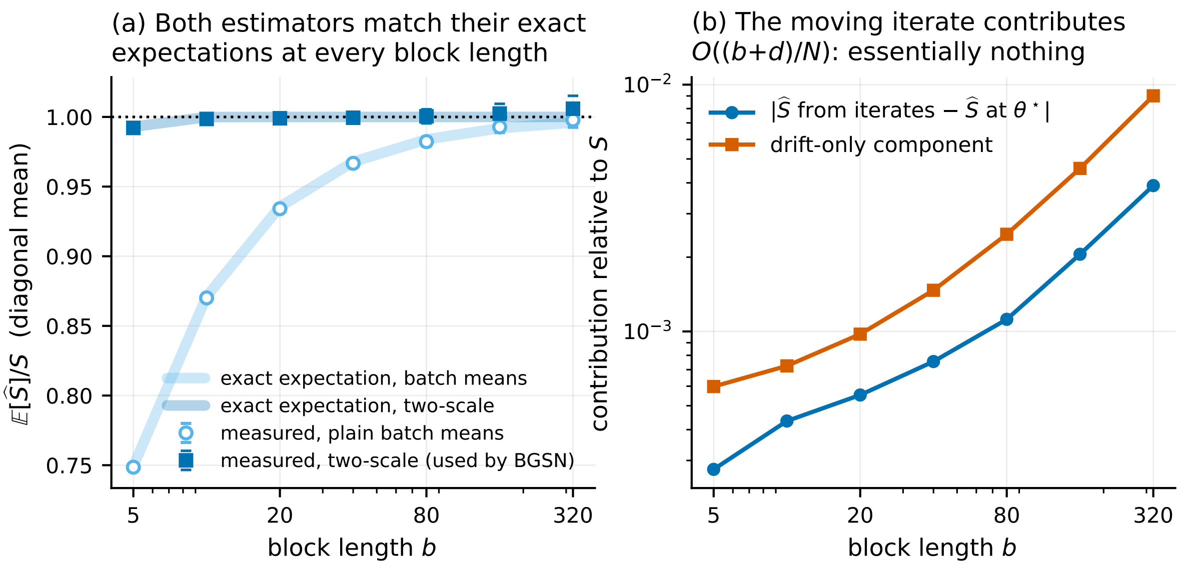

Figure 2: The long-run covariance estimator, and why the moving iterate does not spoil it. (a) Measured $\mathbb{E}[\widehat{S}]/S$ against block length: plain batch means follow the closed-form bias $1 - \Delta/(bS)$ (no fitted constants), while the two-scale form is unbiased at every b, as Theorem 10 predicts under geometric mixing. (b) The algorithm's block gradients decomposed exactly into the score at $\theta^\star$ and the drift $\widehat{H}_t(\theta_{t-1} - \theta^\star)$. The drift's contribution to $\widehat{S}$ is 0.1% of S at $b = 20$ and 0.9% at $b = 320$, growing like $(b + d)/N$ as Theorem 10 predicts, and the estimator built from the algorithm's iterates is indistinguishable from the one built at $\theta^\star$. This is why our block length need only grow like $\log N$ rather than like a power of N (Remark 3).

5.1 The failure is exactly as predicted, and it is not small

Figure 1(a) sweeps the dependence strength and compares the measured coverage of dependence-blind intervals with the parameter-free prediction $2\Phi(z_{0.025}/\sqrt{\kappa}) - 1$. For κ up to 2.9 the largest absolute discrepancy is 0.0063, of the same order as the Monte Carlo standard error: at $\kappa = 2.92$ the prediction is 0.748 and the measurement is 0.742 ± 0.0114. BGSN stays at nominal over that range (0.949 at $\kappa = 2.92$), including at $\phi = \psi = 0$, where all four intervals agree — the control showing that the machinery is not simply widening intervals.

At the strongest dependence we tried, $\kappa = 6.2$, the measurement is 0.560 against a predicted 0.569, and the largest discrepancy over the whole sweep is 0.0085. That gap is finite-sample, not a failure of the formula, and the infeasible oracle interval says so: it covers only 0.946 there, so at $N = 200{,}000$ the central limit theorem itself is not yet accurate at that dependence level. BGSN covers 0.945, i.e. it tracks the oracle rather than the formula — which is the right diagnosis of where the remaining error lives.

The oracle comparison is worth stating as a general point, because it separates two things that are easy to confuse. BGSN's coverage tracks the *infeasible oracle-variance* interval — built from the true H and the true S — to within Monte Carlo error at every dependence level in the sweep. Whatever undercoverage remains is therefore attributable to the central limit theorem at finite N, not to our variance estimator. Plain batch means, by contrast, fall 1–3 percentage points *below* the oracle once $\kappa \geq 4$, which is exactly the $O(1/b)$ bias the two-scale correction removes.

Table 1 gives the full picture. On AR-hom, the base method's interval covers 0.743 and our interval covers 0.949 against a nominal 0.95, with the infeasible oracle interval at 0.950 (which is the sanity check that the central limit theorem itself, and our Monte Carlo harness, are right). The price is width: our interval

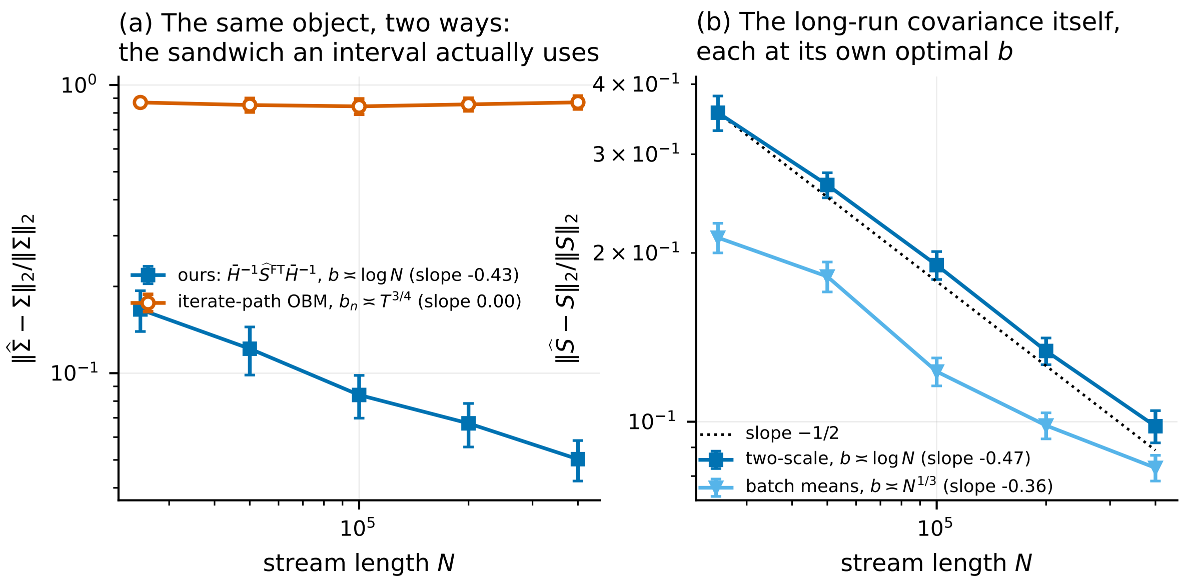

Figure 3: The rate claim, measured, on the object an interval actually uses. Both axes logarithmic; error bars are ±1.96 Monte Carlo standard errors; fitted slopes in the legends. **(a)** Relative operator-norm error in estimating the whole sandwich $\Sigma = H^{-1}SH^{-1}$. Ours is $\bar{H}^{-1}\widehat{S}^{\mathrm{FT}}\bar{H}^{-1}$ at $b \asymp \log N$; the comparator is overlapping batch means on the *iterate* path at the block length $b_n \asymp T^{3/4}$ that analysis requires (Appendix B.3 gives the form), which uses no gradients and no curvature estimate. Same target, so the comparison is meaningful; but read the levels, not the slopes. The iterate-path curve is flat here, and its published $N^{-1/8}$ rate is too slow for our range to resolve, so its fitted slope says nothing about it. What the panel shows is that over the sample sizes we can reach, one estimator is accurate and the other is not. **(b)** Error in estimating S alone, each estimator at its own optimal block length: the two-scale form at $b \asymp \log N$ against the $-1/2$ of Theorem 10(iv), plain batch means at $b \asymp N^{1/3}$ against $-1/3$.

is 1.71× the dependence-blind one, which is exactly $\sqrt{\kappa}$ — there is no free lunch, the dependence-blind interval is simply the wrong length. On the Logistic design, where κ ranges over $[2.27, 2.32]$, the pattern is the same. AR-het is worth a separate look, because its κ_j varies across coordinates by construction: the dependence-blind interval's *per-coordinate* coverage there ranges from 0.63 to 0.96, while ours ranges from 0.92 to 0.98. No scalar inflation of the interval width could produce the first pattern from the second; the whole matrix has to be estimated. Crucially, replacing *only* the variance estimator repairs the base method (row "with our long-run variance"), and replacing only the point estimator does not: this isolates the inference component as the operative one.

The Markov and Logistic columns need a caveat of their own, and it is the sharpest test of whether our variance estimator is really doing the work we claim. Neither design's point estimate has reached its asymptotic regime at $N = 200{,}000$, for two different reasons. The Markov design's regime chain has an integrated autocorrelation time of 66 observations — the sum $1 + 2\sum_{k\geq 1}$ of its autocorrelations, which is the factor by which dependence multiplies the variance of a sample mean, so that N dependent observations carry about as much information as $N/66$ independent ones — so its effective sample size is a small fraction of N and the point estimate is 1.33 times the efficient standard deviation. On the Logistic design the cause is the curvature rather than the dependence: the weight $\alpha = \sigma'(x^\top\theta)$ is largest at $\theta_0 = 0$, where the warm-up evaluates it, so $\bar{H}$ built there overestimates H and the early steps are correspondingly too short; the point estimate is 1.33 times efficient and the fitted standard error is biased low with it. An interval built from the *correct* asymptotic variance must undercover in that situation, no matter who supplied the variance, simply because $\sqrt{N}(\theta_T - \theta^\star)$ is not yet close to its limit. So on these two designs

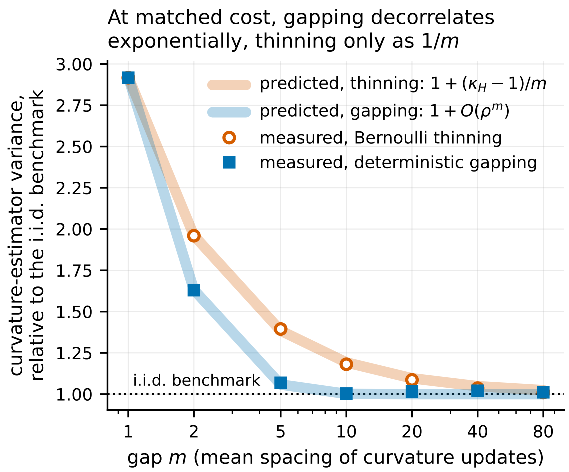

Figure 4: At matched cost, a deterministic gap decorrelates curvature updates exponentially in m while random thinning only does so linearly (Proposition 11). Vertical axis: $n\,\mathbb{E}\|\bar{H}-H\|_F^2$ divided by its exact independent-data value $(\operatorname{tr}\Sigma)^2 + \|\Sigma\|_F^2$ — a computed constant, not a fitted normalisation. Pale bands are the two predictions; markers are measurements with the optimisation frozen at $\theta^\star$, so only estimation quality is compared. $\kappa_H = (1+\phi^2)/(1-\phi^2)$ here.

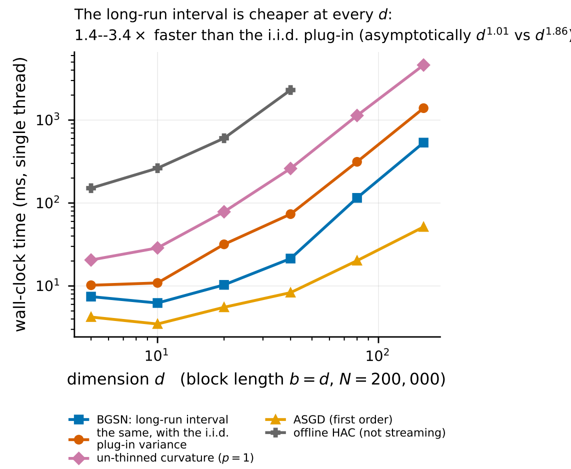

Figure 5: The statistically correct interval is the cheaper one. Wall-clock milliseconds for one pass over $N = 200{,}000$ observations (vertical axis, log scale) against dimension d (horizontal, log scale), single-threaded, minimum over seven interleaved repetitions. The i.i.d. plug-in score covariance costs $O(d^2)$ per *observation*; the blocked long-run covariance costs $O(d^2)$ per *block*. Slopes are fitted on the four largest d; at this N they are inflated by the one-off warm-up (see (9) and Table 6, which also gives the asymptotic exponents), and the two smallest d are dominated by fixed per-block overhead rather than by either term of (9). The message of the figure is the vertical separation, which holds at every d.

the question is not whether coverage is nominal but whether our interval matches the interval that uses the exact $H^{-1}SH^{-1}$ at the same iterate. Table 2 and Figure 8 answer it: over a sweep in N both rise towards nominal together while the dependence-blind interval does not. On the Markov design the agreement is essentially exact: our coverage is within 0.017 of the oracle's at every N, and at the top of the sweep, $N = 500{,}000$, both are 0.918 and 0.917 with the point estimate down to 1.12 times efficient, against 0.494 for the dependence-blind interval. The direction of travel is the whole point. Across the fivefold increase in N our coverage and the oracle's move together towards 0.95 and stay within 0.017 of each other, while the dependence-blind interval creeps up by a few hundredths towards its own asymptote, which Proposition 8 puts at $2\Phi(z_{0.025}/\sqrt{7.55}) - 1$, far below nominal. We stop the sweep at $N = 500{,}000$ rather than continuing to where coverage is nominal, because what the experiment is for is the gap between the first two columns, and Table 2 shows that gap is already at the level of the Monte Carlo standard error there. On the Logistic design there is a real gap, and it is ours: our coverage is 0.736 against the oracle's 0.852 at $N = 100{,}000$, a difference of 0.116, falling monotonically to 0.036 at $N = 500{,}000$, the top of the sweep. The cause is visible in Table 1: our fitted width there is 0.84 times the oracle's, i.e. the standard error is biased low, because $\bar{H}$ is an average over the whole stream and the early blocks contribute $\alpha = \sigma'(0) = 1/4$, its largest possible value, so $\bar{H}$ overestimates H and $\bar{H}^{-1}S\bar{H}^{-1}$ underestimates the sandwich. That bias decays with the Cesàro average of $\|\theta_t - \theta^\star\|$, which is why the gap closes; it is a finite-sample property of averaging curvature along the path, it affects the base method identically, and no

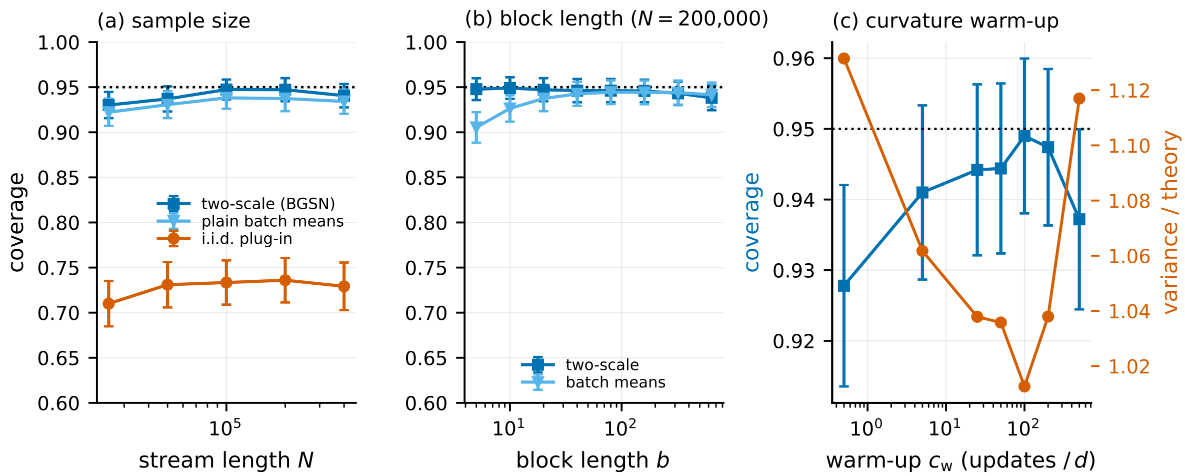

Figure 6: Ablations. (a) Coverage converges to nominal in N for the two-scale variance, while the dependence-blind interval converges to 0.749, the value Proposition 8 predicts for this design. (b) Coverage is flat in the block length for the two-scale variance and degrades at both ends for plain batch means — the practical content of the $O(\rho^b)$-versus-$O(1/b)$ distinction. (c) The curvature warm-up matters, though the step-length safeguard of (6) now absorbs part of its job: with essentially no warm-up coverage on this design is 0.928 against 0.949 at the default, and the relative variance on the right axis tells the same story. On the regime-switching design, where the warm-up is load-bearing rather than merely helpful, the effect is far larger (Table 5(c′)).

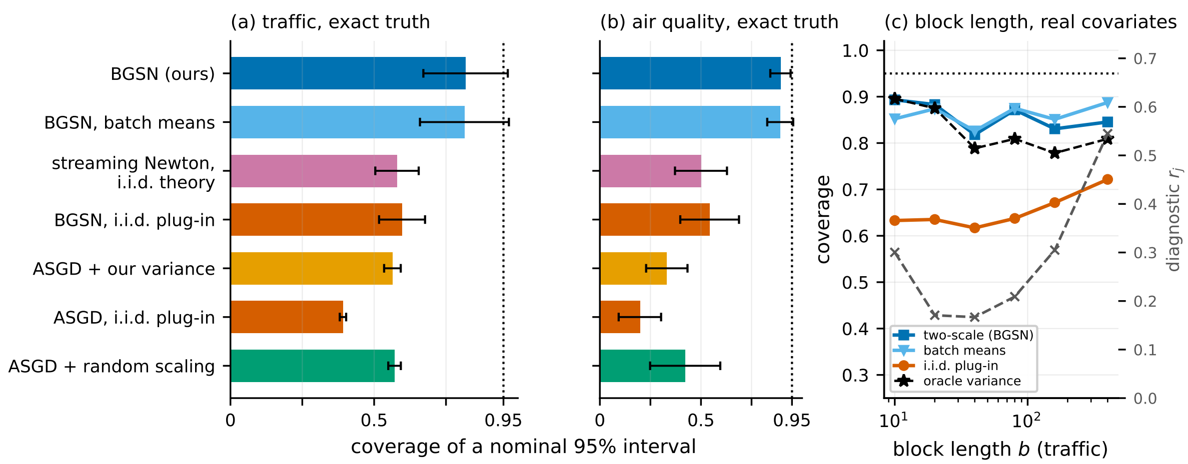

Figure 7: Two real hourly streams. (a, b) Coverage of nominal 95% intervals under Protocol A (real covariates, synthetic autoregressive errors, exact target and exact conditional long-run covariance). The air-quality stream's median estimated variance inflation is 9.9, so dependence-blind intervals fail badly there. (c) Coverage (left axis) against block length b (horizontal, log scale) on the traffic stream, with the free block-adequacy diagnostic $r_j = |u_j^\top(\widehat{S}^{\mathrm{FT}} - \widehat{S}^{\mathrm{BM}})u_j| / (u_j^\top \widehat{S}^{\mathrm{BM}} u_j)$ — an estimate of the relative block-length bias of plain batch means, defined where it is first used in Section 5 — on the right axis. Three things to read. Our coverage is flat in b over a fortyfold range and stays within a few hundredths of the infeasible oracle-variance curve throughout, so the shortfall from 0.95 on an 8,000-observation segment is the point estimate and not the covariance. The dependence-blind curve sits far below at every b and does not close. And r_j is U-shaped, lowest near $b = 20$–40: it rises for small b, where plain batch means are biased, and again for large b, where too few blocks remain to estimate anything stably — which is what makes it a usable guide to the block length, though not a calibration guarantee.

choice of long-run covariance estimator would remove it. That is the signature we want: our error tracks the oracle's at every sample size, and the dependence-blind error is the one that survives $N \to \infty$. The step-length safeguard bound at most 8 times per run anywhere in that sweep, and the count does not rise with N, which is the empirical content of Proposition 1.

We report every design in Table 1 at the same N rather than tuning N per design, which is why the Markov column looks worst there; Table 2 is where that column is resolved.

The step-length safeguard is what makes the hard design run at all. Table 5(i) varies the safeguard. With none, the autoregressive designs are barely affected (1.05 against 1.04 in relative variance) but the regime-switching design diverges: relative variance 1.1×10^{14} against 1.41 with the safeguard, and coverage 0.228 against 0.907. The safeguard is not doing this by damping the algorithm into submission: it binds a median of 7 times in 10,000 blocks on that design and 3 times on AR-hom, and Table 2 shows the count flat as N grows by a factor of five.

The constant c_D behaves the way a safeguard constant should, which is to say it is irrelevant until it is not. On AR-hom, changing it by a factor of four in either direction moves nothing (1.04 at $c_D = 1/4$ and 1.04 at $c_D = 4$, against 1.04 at our default $c_D = 1$). On the design that actually needs the safeguard both directions cost something, and asymmetrically: $c_D = 1/4$ gives 1.59 against 1.41 — binding 21 times instead of 7, it now restrains steps that were fine — while $c_D = 4$ is too loose to prevent the blow-up at all and the relative variance is 4.3×10^7. So c_D is a tuning constant with a real failure mode on one side, and we say so rather than advertising insensitivity we only observe on the easy designs.

The schedule shift of Appendix C.1 is the alternative the contraction condition suggests more directly, and it is measurably worse: 3.08 against 1.41 on the regime-switching design, because it slows all 10^4 steps to correct a handful. We report it because we implemented it first, and because the comparison is the evidence for the design we kept.

The first-order rows of Table 1 need a caveat, and it cuts against us if we hide it. These designs are ill-conditioned — $\operatorname{cond}(H) = 400$ for AR-hom — which is precisely the regime streaming second-order methods exist for, and there averaged SGD's *point* estimate has not converged: its root-mean-square error is 9.77 times the efficient value against 1.01 for BGSN. Its intervals are therefore uninformative about the inference question, whatever covariance they use. Two pieces of evidence show that this is about conditioning and not about our construction. First, within Table 1 itself: streaming AdaGrad, which rescales per coordinate, is efficient on AR-het (1.03) and with our long-run variance it covers 0.947 — so the construction transfers to a first-order method as soon as that method's point estimate converges. Second, Table 3 repeats the AR-hom design with $\Sigma_x = I$, where every point estimator is efficient (1.07 for averaged SGD, 1.03 for BGSN) and only the covariance can decide coverage: averaged SGD with our long-run variance covers 0.931, the same estimator with the i.i.d. plug-in covers 0.721, and BGSN covers 0.942.

Coverage checks a single quantile, while the theorem claims a whole distribution, so we also report the Kolmogorov–Smirnov distance between the studentised error (the error divided by its own estimated standard error, which is what an interval implicitly compares to a normal quantile) $\sqrt{N}(\theta_{T,j} - \theta^\star_j)/\{u_j^\top \widehat{S}^{\mathrm{FT}} u_j\}^{1/2}$ and the standard normal, taken per coordinate across replicates. On AR-hom the worst of the d coordinates gives 0.089 against a 5% critical value of 0.079: marginally outside, so at this number of replicates the studentised distribution is distinguishable from normal, and we report that rather than rounding it in our favour. The dependence-blind studentisation gives 0.207, an order of magnitude further out. The comparison, not the test outcome, is the content: our studentisation is at the edge of detectable non-normality while the dependence-blind one is not close. The standard errors themselves are accurate: 0.997 times the truth on average.

5.2 The long-run covariance estimator behaves as the theory says

Figure 2(a) plots $\mathbb{E}[\widehat{S}]/S$ against the block length against the *exact* prediction $\sum_{|k|<b}(1 - |k|/b)\Gamma_k/S$, which is closed form for this design (we plot the exact Bartlett-weighted sum rather than its $1/b$ expansion, so the comparison carries no approximation of its own). Plain batch means track it over the whole range, at $b = 20$ measured 0.934 against predicted 0.936, and the two-scale estimator is within Monte Carlo error of 1 for $b \geq 10$ (at $b = 20$: 0.999); the small residual at $b = 5$ is the $O(\rho^b)$ term, which is only $\rho^5 = 0.03$ there. This is Theorem 10(i)–(ii) with no fitted constants.

Figure 3 measures the rate. Panel (b) is the rate for S itself: under the $b \asymp \log N$ schedule the two-scale estimator's relative error falls with fitted exponent -0.47 against the $-1/2$ of Theorem 10(iv), while plain batch means under their own optimal $b \asymp N^{1/3}$ schedule give -0.36 against $-1/3$.

Panel (a) puts our estimator and an iterate-path estimator on the same object, because an interval uses the whole sandwich $\Sigma = H^{-1}SH^{-1}$ rather than S alone. We compare our composite $\bar{H}^{-1}\widehat{S}^{\mathrm{FT}}\bar{H}^{-1}$, whose relative error falls with fitted exponent -0.43, with overlapping batch means on the *iterate* path at the block length its analysis prescribes — the estimator of Σ that uses no gradients and no curvature.

Two things must be said carefully here, because the comparison is easy to overstate. First, the iterate-path estimator's relative error is flat over our range: the fitted exponent is 0.00, and we do *not* read that as its rate. Its published rate is $N^{-1/8}$, which over a sixteenfold increase in N predicts a reduction by a factor $16^{-1/8} = 0.71$ — a change small enough that our range cannot separate it from zero, and our fitted exponent should be treated as uninformative about it rather than as a measurement. Second, what our range *does* establish is the level, and the level is the practical statement: over the whole range the iterate-path estimator's relative error in Σ sits near 0.86, which is not a usable standard error whatever its asymptotic rate, while ours falls from 0.17 to 0.05. We are comparing an estimator that has entered its asymptotic regime with one that has not, and the honest claim is about the sample sizes a practitioner would actually have, not about which exponent is larger.

Figure 2(b) addresses the objection that a block average must span enough parameter movement to behave like an independent draw — the reason iterate-based batch means need $b \asymp N^{3/4}$ [83]. We decompose the algorithm's block gradients exactly into the score at $\theta^\star$ and the drift $\widehat{H}_t(\theta_{t-1} - \theta^\star)$ and evaluate the long-run covariance of each. The drift component contributes 0.0010 of S at $b = 20$ and grows like $\tau_{b,N} = (b + d)\log^2 N/N$, confirming Remark 3: because the blocks carry gradients rather than iterates, the block length is set by the data's mixing time, not the optimiser's.

5.3 Curvature schedules

Figure 4 compares deterministic gapping with Bernoulli thinning at a matched number of rank-one updates, with the optimisation frozen at $\theta^\star$ so that only estimation quality is measured. The excess variance over the independent-data benchmark follows the two predicted curves — $O(\rho^m)$ for gapping, $(\kappa_H - 1)/m$ for thinning — over more than an order of magnitude in m. As stated in Section 7, this is a curvature-side result: Table 5(f) shows that the end-to-end effect on coverage is within Monte Carlo error, which is what the theory predicts, since $\bar{H}$ enters the limit only through a rate.

Where gapping does pay is cost, and Table 1 measures it. Compare BGSN with the row "base method with our long-run variance": the two use the same variance estimator and the same number of rank-one curvature updates per observation ($1/d$), and they cover the same (0.949 against 0.949 on AR-hom). They differ only in *how* those updates are placed — on a deterministic gap of $m = d$ against Bernoulli retention with $p = 1/d$. At these settings the wall-clock difference is 1.04×, which is within our timing noise, so we do not claim a speed-up: a gap avoids drawing and testing a random variable per observation, but that saving is small next to the $O(d^2)$ update it guards. The case for gapping is therefore the one Proposition 11 makes and Figure 4 measures — a strictly better curvature estimate at a matched number of updates, $O(\rho^m)$ excess variance against $(\kappa_H - 1)/m$ — together with the fact that it makes the per-block cost deterministic rather than random. It is a better cost knob, not a faster one.

5.4 Ablations

Figure 6 and Table 5 vary one design choice at a time. (a) Coverage rises monotonically to nominal with N, and the gap to the dependence-blind interval does *not* close — it converges to the predicted 0.749. (b) Coverage as a function of b is flat for the two-scale estimator over $b \in [10, 320]$ and degrades at both ends for plain batch means (bias at small b, noise at large b): the practical content of the $O(\rho^b)$-versus-$O(1/b)$ distinction. (c) The curvature warm-up matters, though less than it did before the step-length safeguard absorbed part of its job: with essentially no warm-up, coverage on this design is 0.928 against 0.949 at the default, and above $c_w \approx 25$ the result is insensitive. (c′) is where the warm-up is load-bearing: on the regime-switching design the relative variance is 2.19 at $c_w = 50$ against 1.41 at $c_w = 200$, because there $50d$ consecutive observations span only about thirty regime visits and leave the curvature estimate rank-starved. That is what sets our default, and it is a statement about effectively distinct design points rather than about observations. (d) Results are insensitive to the ridge exponent q_r over the whole admissible range, and to the ridge constant c_ι from 0 — no ridge at all — up to our default 0.01; at $c_\iota = 0.1$ coverage slips by two points and at $c_\iota = 1$ the ridge dominates the data terms and the estimator is no longer efficient. So the ridge is safe to leave alone but not safe to enlarge, and the honest reading is that the default sits at the top of the flat region rather than in the middle of it. That $c_\iota = 0$ works as well as the default is worth stating plainly: the ridge is in the algorithm so that Lemma 2 can hold for *every t* without an assumption on the iterates, not because the runs need it. (e) Results are insensitive to the initial curvature scale λ_0 over eight orders of magnitude, which matters because that is a parameter a user would otherwise have to think about. (f) Gap and thinning behave as Proposition 11 predicts and do not move coverage, which is what the theory says: $\bar{H}$ enters the limit only through a rate. (g) At strong dependence ($\phi = \psi = 0.85$, $\kappa = 6.2$) the two estimators separate in exactly the way the $O(\rho^b)$-versus-$O(1/b)$ distinction predicts. Plain batch means need a long block: coverage rises from 0.921 at $b = 20$ to 0.946 at $b = 160$. The two-scale correction does not: it is already at 0.944 at $b = 20$ and stays there (0.945 at $b = 160$). The free diagnostic r_j falls from 0.177 to 0.086 over the same range, which is what makes it usable as a guide to whether plain batch means would have been adequate. So the correction is what buys the short block, and the short block is what keeps the estimator cheap. (h) The weighted averaged variant is slightly better at small N and slightly worse at large N; both are reported. The non-positive-definiteness fallback of Remark 4 never triggered anywhere: the rate is 0 in every simulation and 0.0% of coordinate-replicate pairs on the real streams, where an earlier version of these experiments did not record it at all and where we would have expected it if anywhere, since there the lag-one correction is largest. It never triggered (0.0% of all runs, all panels).

5.5 The correct interval is also the cheaper one

The authoritative statement is the operation count, because it is exact and independent of the machine. The streaming pass grows as $d^{1.00}$ for BGSN and as $d^{1.86}$ once the i.i.d. plug-in variance is added, because the plug-in score covariance costs $O(d^2)$ per *observation* whereas our long-run covariance costs $O(d^2)$ per *block*; evaluating the whole count at $N = 10^8$ gives $d^{1.01}$ against $d^{1.86}$. Those are the design claim, and they are arithmetic rather than a fit.

The exponents fitted at the N of Table 6 are *not* the streaming cost, and we say so rather than quoting them as if they were: with $c_w = 200$ and $N = 200{,}000$ the one-off $O(c_w d^3)$ warm-up dominates the $O(dN)$ pass for $d \geq 40$ (the crossover is $N \gtrsim c_w d^2$), so both measured exponents sit near 2. What is robust at this N is the *level*: the plug-in variant costs more than BGSN in wall-clock at every d we tried, by a factor between 1.4 and 3.4 (median 2.7), and in the main configuration BGSN runs in 0.37 times the time of the dependence-blind streaming Newton method — that is, our valid intervals cost less than the invalid ones, not more. The measured wall-clock exponents, on a shared and heavily loaded machine, are $d^{1.95}$ and $d^{1.85}$; they are noisier than the counts and we report them only as corroboration. Figure 5 plots the same timings against d, where the message is the vertical separation rather than either slope.

5.6 Real streams

Table 4 reports two protocols on hourly traffic volume on Interstate 94 and hourly PM2.5 at twelve Beijing monitoring sites. After a fixed regression adjustment (weather plus calendar harmonics) the residual lag-one autocorrelation is 0.76 and 0.91 respectively, and the median estimated variance inflation across coordinates is 5.8 for traffic and 9.9 for air quality. Table 4 reports, for each stream, the pooled prediction $d^{-1}\sum_j\{2\Phi(z_{0.025}/\sqrt{\kappa_j})-1\}$ next to the measurement, so the comparison is against the quantity the table actually measures rather than a single representative κ.

Protocol A keeps the real (strongly dependent) covariate stream and replaces the response by $x_t^\top\theta^\star+u_t$ with u an autoregression whose coefficient equals that of the real residuals. This gives an *exact* target and an exact conditional long-run covariance while retaining real covariate dependence: BGSN covers 0.818 (traffic) and 0.893 (air quality) against nominal 0.95, while the dependence-blind interval covers 0.597 and 0.543. Protocol B uses the real response and takes the full-stream M-estimator as the target; because that target is itself estimated from a superset of each segment, the correct nominal level for the comparison is $2\Phi(z_{0.025}/\sqrt{1-1/R})-1$, which we compute exactly and report in the table header. BGSN covers 0.795 and 0.858. Neither protocol alone would settle the matter—A has an exact target but a synthetic error process, B has a real error process but an estimated target—and they agree.

Figure 7(c) shows why the block length is the one parameter a practitioner must think about on real data, and that the free diagnostic $r_j=|u_j^\top(\widehat{S}^{\mathrm{FT}}-\widehat{S}^{\mathrm{BM}})u_j|/(u_j^\top\widehat{S}^{\mathrm{BM}}u_j)$ flags the problem. The denominator is the batch-means quadratic form, not the two-scale one: $\widehat{S}^{\mathrm{BM}}$ is a sum of outer products and so its quadratic forms are non-negative, whereas those of $\widehat{S}^{\mathrm{FT}}$ can be negative (Remark 4), and an earlier version of this diagnostic divided by the latter and produced meaningless numbers on the real streams. On the traffic stream r_j is 0.16 at the $b=40$ we use, and the sweep in Figure 7(c) shows what it is good for: r_j is U-shaped in b, rising both for short blocks, where plain batch means carry the $O(1/b)$ bias, and for long ones, where too few blocks remain for a stable estimate. Our coverage over that fortyfold range in b is flat and stays within a few hundredths of the infeasible oracle-variance curve, which is the evidence that the shortfall from nominal on these segments is the point estimate rather than the covariance. So r_j is a guide to choosing b, not a calibration guarantee, and we do not claim it predicts coverage.

6 Related work

This paper sits at the intersection of three literatures, and the honest statement of what is new requires being precise about each.

Streaming second-order methods with $O(dN)$ cost. Inversion-free stochastic Newton methods maintain a running inverse curvature estimate by rank-one Sherman–Morrison updates [84] and have been developed for logistic regression [6], Gauss–Newton problems [12], the geometric median [33], and general conditioned stochastic gradient descent [9, 55]; Chau et al. [13] survey the inversion-free idea more broadly, and Byrd et al. [11], Moritz et al. [65], Wang et al. [93] give the stochastic quasi-Newton and L-BFGS

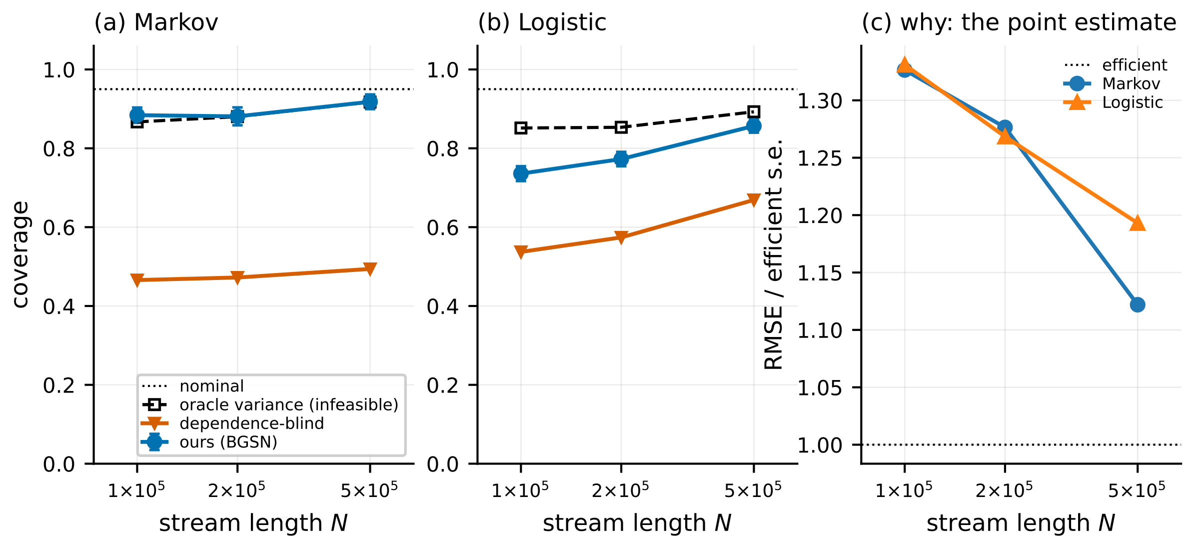

Figure 8: Separating "the variance estimate is wrong" from "the central limit theorem has not taken hold" on the two designs whose point estimate has not converged at the sample sizes of Table 1. **(a)** The regime-switching design, whose covariate chain has an integrated autocorrelation time of about 66 observations. **(b)** The dependent-logistic design, where instead the curvature weight $\alpha = \sigma'(x^\top\theta)$ is largest at the θ_0 the warm-up evaluates it at. In both, the infeasible curve uses the exact $H^{-1}SH^{-1}$ at the same iterate, so the vertical distance between it and ours is everything our covariance estimator is responsible for: that distance is at most 0.017 on the Markov design and falls from 0.116 to 0.036 on the logistic one. Ours and the infeasible curve rise towards nominal together; the dependence-blind curve does not. **(c)** Why both left panels start low: the root-mean-square error of the point estimate, divided by the efficient standard error, is still above one, so $\sqrt{N}(\theta_T - \theta^\star)$ is not yet close to its limit and no choice of variance can give nominal coverage.

Table 2: On the two designs whose point estimate has not converged, our interval tracks the infeasible oracle interval at every sample size ($d = 20$, $b = 20$; R decreases from 200 at the smallest N to 150 at the largest, since what is being measured is the gap between the first two columns and a Monte Carlo standard error near 0.01 is ample for it). *ours* is BGSN; *oracle* uses the exact $H^{-1}SH^{-1}$ at the same iterate and is infeasible; *blind* uses the i.i.d. plug-in. *err* is the root-mean-square error divided by the efficient standard error, so err = 1 means the point estimate has reached the asymptotic regime, and *clips* counts activations of the step-length safeguard out of N/b blocks. At $N = 200{,}000$ neither design is nominal because err > 1: the central limit theorem has not taken hold, which no variance estimator can repair. What our estimator is responsible for is the difference between the *ours* and *oracle* columns, and the largest such difference over both designs and all four sample sizes is 0.116. The *blind* column is the one that does not converge. Note that *clips* does not grow with N, which is the empirical content of Proposition 1.

design	N	ours	oracle	blind	err	clips
Markov	100,000	0.884	0.867	0.466	1.33	8
	200,000	0.881	0.880	0.472	1.28	8
	500,000	0.918	0.917	0.494	1.12	7
Logistic	100,000	0.736	0.852	0.537	1.33	2
	200,000	0.773	0.853	0.574	1.27	2
	500,000	0.857	0.893	0.669	1.19	2

Table 3: A well-conditioned dependent design isolates the inference question ($\Sigma_x = I$, so $\mathrm{cond}(H) = 1.00$; otherwise identical to the AR-hom column of Table 1: $d = 20$, $N = 200{,}000$, $b = 20$, $\kappa = 2.92$, $R = 300$). Here every point estimator is efficient ($err \approx 1$), so coverage is decided by the covariance estimate alone. Averaged SGD with our long-run variance is calibrated; the same estimator with the i.i.d. plug-in is not, by the factor $\sqrt{\kappa} = 1.71$ in width that Proposition 8 predicts. On the ill-conditioned designs of Table 1 the first-order baselines cannot be read this way, because there their point estimates have not converged.

method	cov	w	err
BGSN (this paper)	**0.942**	1.00	1.03
with plain batch means	0.933	0.97	1.03
with the i.i.d. plug-in variance	0.743	0.59	1.03
streaming Newton, i.i.d. theory	0.740	0.59	1.03
ASGD + our long-run variance	0.931	1.00	1.07
ASGD, i.i.d. plug-in	0.721	0.59	1.07
ASGD + random scaling	0.917	1.28	1.07
AdaGrad + our long-run variance	**0.943**	1.00	1.03
oracle variance *(infeasible)*	**0.941**	1.00	1.03

Table 4: Real hourly streams. Protocol A: real covariates, synthetic autoregressive errors whose coefficient equals the AR(1) coefficient of the real residuals, so the target and the long-run covariance are exact conditionally on the covariate path. Protocol B: fully real response, target the full-stream M-estimator; because that target is estimated from a superset of each segment, the correct nominal level for this comparison is $2\Phi(z_{0.025}/\sqrt{1 - 1/R}) - 1$, reported in the header. Traffic: $n = 8000$ per segment, $b = 40$. Air quality: $n = 10000$ per segment, $b = 50$, pooled over 12 monitoring sites.

	traffic (I-94)		air quality (PM2.5)	
method	A: exact truth	B: full-stream target	A: exact truth	B: full-stream target
nominal	0.950	0.976	0.950	0.994
BGSN (this paper)	0.818	0.795	0.893	0.858
with plain batch means	0.814	0.795	0.892	0.816
with the i.i.d. plug-in variance	0.597	0.682	0.543	0.361
streaming Newton, i.i.d. theory	0.580	0.682	0.500	0.365
ASGD + our long-run variance	0.564	0.523	0.331	0.330
ASGD, i.i.d. plug-in	0.391	0.455	0.198	0.212
ASGD + random scaling	0.571	0.568	0.422	0.486
oracle variance *(infeasible)*	0.773	–	0.788	–
stream diagnostics				
residual AR(1) coefficient	0.756		0.909 (median over sites)	
R^2	0.782		0.327	
block-adequacy r_j	0.163	0.271	0.186	0.302

Table 5: Ablations on the homogeneous autoregressive design ($d = 20$, $\kappa = 2.92$, $R = 250$). *cov* columns are coverage of nominal 95% intervals using the two-scale, plain batch-means, and i.i.d. plug-in variance respectively; *eff* is the empirical variance of the point estimate divided by its asymptotic value (1.00 is efficient); *adeq* is the free block-adequacy diagnostic r_j, an estimate of the relative block-length bias of plain batch means. Panel (c′) is on the regime-switching design and is the reason the default warm-up is $c_w = 200$ rather than 50: there, $50d$ consecutive observations span only about thirty regime visits, the curvature estimate is still rank-starved, and the $1/t$ schedule diverges. The warm-up must span enough effectively distinct design points, which is not the same as enough observations. Panel (i) varies how the blocked step is safeguarded: *none* is the raw $1/t$ step, *cap* is the step-length safeguard of (6) at $c_D = 1$ (our default) with *cap 0.25* and *cap 4* changing c_D by a factor of four either way, *shift* is the rejected schedule shift of Appendix C.1, and *both* applies the two together; *clips* counts how many of the 10,000 blocks the safeguard acted on. *eff* is in scientific notation where the unsafeguarded step diverges. Panel (j) is on a short stream ($N = 4000$, $d = 11$, the length of a real-data evaluation segment) and varies the cap on the warm-up as a fraction of the stream; $\varpi_w = 1$ means uncapped, which spends 2200 of the 4000 observations on curvature and leaves 45 blocked steps.

panel	setting	eff	cov (2-scale)	cov (BM)	cov (plug-in)	adeq
(a) length N	$N = 25{,}000$	1.23	0.930	0.922	0.710	0.086
	$N = 50{,}000$	1.08	0.937	0.931	0.731	0.072
	$N = 100{,}000$	1.04	0.947	0.938	0.733	0.069
	$N = 200{,}000$	1.04	0.947	0.937	0.736	0.069
	$N = 400{,}000$	1.10	0.941	0.934	0.729	0.069
(b) block b	$b = 5$	1.02	0.948	0.905	0.736	0.325
	$b = 10$	1.03	0.949	0.926	0.737	0.149
	$b = 20$	1.04	0.947	0.937	0.736	0.069
	$b = 40$	1.04	0.946	0.943	0.737	0.041
	$b = 80$	1.05	0.946	0.944	0.738	0.045
	$b = 160$	1.06	0.946	0.944	0.734	0.060
	$b = 320$	1.07	0.943	0.944	0.738	0.085
	$b = 640$	1.10	0.938	0.942	0.733	0.118
(c) warm-up	$c_w = 0.5$	1.13	0.928	0.921	0.724	0.069
	$c_w = 5$	1.06	0.941	0.931	0.730	0.069
	$c_w = 25$	1.04	0.944	0.936	0.737	0.068
	$c_w = 50$	1.04	0.944	0.935	0.741	0.068
	$c_w = 100$	1.01	0.949	0.941	0.739	0.069
	$c_w = 200$	1.04	0.947	0.938	0.733	0.069
	$c_w = 500$	1.12	0.937	0.927	0.713	0.069
(c′) warm-up, Markov	$c_w = 50$	2.19	0.848	0.812	0.452	0.239
	$c_w = 100$	1.76	0.879	0.851	0.467	0.238
	$c_w = 200$	1.41	0.907	0.877	0.495	0.237
	$c_w = 500$	1.21	0.923	0.900	0.536	0.235
(d) ridge	$c_\iota = 0,\ q_r = 0.2$	1.04	0.949	0.940	0.741	0.069
	$c_\iota = 0.001,\ q_r = 0.2$	1.04	0.949	0.940	0.740	0.069
	$c_\iota = 0.01,\ q_r = 0.2$	1.04	0.947	0.937	0.736	0.069
	$c_\iota = 0.1,\ q_r = 0.2$	1.04	0.931	0.920	0.703	0.069
	$c_\iota = 1,\ q_r = 0.2$	3.07	0.605	0.591	0.405	0.069
	$c_\iota = 0.01,\ q_r = 0.05$	1.03	0.943	0.931	0.729	0.069
	$c_\iota = 0.01,\ q_r = 0.1$	1.03	0.946	0.934	0.732	0.069
	$c_\iota = 0.01,\ q_r = 0.24$	1.04	0.948	0.938	0.737	0.069
	$c_\iota = 0.01,\ q_r = 0.4$	1.04	0.949	0.939	0.740	0.069
	$c_\iota = 0.01,\ q_r = 0.49$	1.04	0.949	0.940	0.740	0.069
(e) H_0 scale	$\lambda_0 = 1e-08$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 1e-06$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 0.0001$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 0.01$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 1$	1.03	0.947	0.935	0.734	0.069
	$m = 1,\ p = 1$	1.02	0.949	0.940	0.740	0.069

Table 6: Cost. Wall-clock milliseconds for one pass over $N = 200{,}000$ observations, single-threaded, minimum over 7 interleaved repetitions (interleaved so that a change in the load of this shared machine cannot be mistaken for a difference between methods; the minimum because load can only make a measurement slower). s_{time} and s_{flop} are the exponents in d fitted on the four largest d to the times and to the exact operation counts at this N. Both are dominated by the one-off $O(c_{\text{w}} d^3)$ warm-up here and should *not* be read as the streaming cost: s_{str} is the same exact count with the warm-up excluded and s_∞ is it at $N = 10^8$, where the warm-up is negligible and (9) reduces to $O(dN)$. Those last two columns are the design claim: the streaming pass is linear in d for BGSN and quadratic once the i.i.d. plug-in variance is added, because the plug-in score covariance costs $O(d^2)$ per *observation* while our long-run covariance costs $O(d^2)$ per *block*. The measured *levels* say the same thing and are robust to load: the plug-in variant costs between 1.4 and 3.4 times more than BGSN, at every d here.

method	d=5	d=10	d=20	d=40	d=80	d=160	s_{time}	s_{flop}	s_{str}	s_∞
BGSN (long-run interval)	7.4	6.2	10.3	21.4	115.4	534.9	1.95	2.07	1.00	1.01
with the i.i.d. plug-in variance	10.2	10.9	31.8	73.5	313.9	1392.3	1.85	2.02	1.86	1.86
streaming Newton, Bernoulli $p = 1/d$	9.4	7.7	11.2	27.0	111.2	533.9	1.88	2.07	–	–
streaming Newton, every observation ($p = 1$)	20.6	28.8	78.4	260.7	1135.1	4608.5	1.98	1.97	1.97	1.97
ASGD (first order, random scaling)	4.2	3.5	5.5	8.3	20.2	51.5	1.09	–	–	–
offline HAC, $L = 100$ *(not streaming)*	151.1	263.2	604.1	2303.0	–	–	–	–	–	–

counterparts. Our starting point is Godichon-Baggioni and Werge [34], who thin the rank-one updates with a Bernoulli variable and take mini-batches of size d to reach $O(dN)$ total cost with asymptotic efficiency, and Godichon-Baggioni et al. [36], who reach the same budget by masking Hessian columns. **The $O(dN)$ budget and the idea of skipping curvature updates are theirs, not ours**; Chen et al. [16] independently thin Hessian *entries* with probability $O(1/d^2)$ in a zeroth-order method. All of this work assumes independent observations. Most of it proves central limit theorems and stops there, but not all: for averaged first-order algorithms under independent data Godichon-Baggioni [32] gives a recursive online estimator of the asymptotic variance with almost sure and L^2 rates, and Chen et al. [15], Godichon-Baggioni and Lu [33], Zhu et al. [103] supply plug-in and batch-means constructions. **Online estimation of the *instantaneous* sandwich under independent data is therefore not new either**; what is missing is the long-run object under dependence, and the streaming second-order setting. Our contribution on this axis is (i) the analysis of the rank-one recursion when the accumulated terms are dependent, (ii) replacing Bernoulli thinning by a deterministic gap, which is strictly better at matched cost (Proposition 11), and (iii) turning the central limit theorem into usable intervals.

Online inference for stochastic approximation. Under independent sampling this is a mature area. Chen et al. [15] give the plug-in and increasing-batch-means variance estimators for averaged stochastic gradient descent; Zhu et al. [103] make batch means fully recursive with growing overlapping windows and obtain $\|\widehat{\Sigma} - \Sigma\| = O(N^{-1/8})$; Singh et al. [85] show equal batch sizes are preferable and apply the flat-top bias correction; Ni and Huo [69] improve the batch-means rate to $N^{-1/6}$ and establish the minimax rate $\Theta(N^{-(1-a)/2})$ for *Hessian-free* covariance estimation from the iterate path; Wei et al. [94] give a fully online de-biased estimator that needs no second-order information and reaches $N^{(a-1)/2}\sqrt{\log N}$, i.e. that minimax rate up to a logarithm, so the Hessian-free class is close to saturated. Fang [26], Fang et al. [27] bootstrap; Su and Zhu [86] split the trajectory; Chen et al. [16], Lee et al. [54], Li et al. [58] studentise by a random scaling matrix and estimate nothing. Leung and Chan [56] and Wu [100] design recursive long-run variance estimators directly. **None of these is a claim we duplicate**; what we take from them is the batch-means template and the flat-top correction, and what we add is the block-*gradient* construction and its consequence for the rate (Remark 3).

Two recent papers give online covariance estimation for streaming *second-order* methods: Kuang et al. [50] for a sketched Newton method and Wang et al. [92] for Nesterov-accelerated sketching. Both are under independent data, both target the instantaneous sandwich, and both require the gradient noise to be a martingale difference with respect to the algorithm's filtration (50, Assumption 3.2) — exactly the condition that temporal dependence destroys. Kuang et al. [50] remark that their analysis extends to conditioned stochastic gradient methods generally, which would include an independent-data version of our algorithm; the martingale assumption is what keeps that remark from covering the present setting.

Stochastic approximation on dependent data. Convergence of stochastic approximation under Markovian or mixing noise is classical [4, 5, 7, 29, 44, 52], and modern rates are available for Markov-chain gradient descent [1, 20, 23, 24, 49, 66, 87]. Godichon-Baggioni et al. [35] analyse streaming stochastic gradient methods with time-dependent and biased gradients, and identify blocked or growing mini-batches as the device that breaks dependence; they give L^2 rates but no central limit theorem and no inference. **Blocked gradients as a dependence-breaking device are theirs.**

On the inference side, Liu et al. [60] give a multiplier bootstrap for ϕ-mixing streams using alternating growing blocks, and establish the long-run sandwich limit and, qualitatively, the failure of dependence-blind intervals; Li et al. [57] give a functional central limit theorem and the semiparametric efficiency bound $H^{-1}SH^{-1}$ under Markovian sampling, with random scaling for inference; Samsonov et al. [83] give Berry–Esseen bounds and an overlapping-batch-means variance for *linear* stochastic approximation with Markovian noise; Roy and Balasubramanian [81] give an online batch-means estimator of the Markovian long-run sandwich; Li et al. [58] treat controlled Markov chains with random scaling. **The long-run sandwich limit, the efficiency bound, and the existence of online long-run covariance estimators under dependence are all prior art** — for first-order methods. Every one of these works is first-order; none maintains a curvature inverse, and the two that estimate a long-run covariance do so from the iterate path, at $O(N^{-1/8})$, with block lengths that must grow like $N^{3/4}$ (83, Prop. 2, Cor. 2). Remark 3 explains why block gradients escape that constraint, and Theorem 10 quantifies the gain.

Long-run variance estimation. The statistical device is heteroskedasticity- and autocorrelation-consistent covariance estimation [2, 3, 67, 68, 95, 97], its fixed-b alternative [47, 48, 88], and the batch-means literature from simulation output analysis and Markov chain Monte Carlo [17, 28, 30, 31, 43, 91]. The two-scale correction we use is the flat-top lag window of Politis and Romano [73], refined by Politis [72] and recast as "lugsail" lag windows by Vats and Flegal [90]; Singh et al. [85] bring it to stochastic gradient descent under independent data, where their Eqs. (15) and (17) coincide with our (13) and their Corollary 1 shows the convergence *rate* is unchanged. **We do not claim the correction.** We claim the bias order it attains at block-gradient resolution under geometric mixing, which is the property that makes $b \asymp \log N$ admissible, and the streaming implementation. Block bootstrap and subsampling [51, 53, 74] are the resampling counterparts.

Asymptotics for dependent M-estimation. The population statements we use are standard: sandwich asymptotics for M-estimators and generalised method of moments [38, 41, 89, 96], misspecified models with dependent observations [21], marginal models for correlated responses [59, 102] — the setting our logistic design imitates — and central limit theorems for mixing sequences [10, 19, 22, 39, 42, 61, 77, 80, 98]. The technical inputs to our proofs are Davydov's covariance inequality [18, 76], the Bernstein-type inequality for weakly dependent sequences of Merlevède et al. [62], the almost sure behaviour of stochastic algorithms [25, 64, 70, 71, 79], and the functional central limit theorem of Herrndorf [39].

7 Limitations

We list the boundaries of the result plainly, because several of them are load-bearing.

Memory is $O(d^2)$, not $O(d)$. The $O(dN)$ claim is about *time*. The method stores a $d \times d$ inverse curvature estimate and two $d \times d$ covariance accumulators, so memory is $O(d^2)$; this is inherited from the independent-data method we extend and is not something we improve. For d in the thousands this is the binding constraint, and a sketched or diagonal-plus-low-rank curvature representation—as in Godichon-Baggioni et al. [36], Kuang et al. [50], Wang et al. [92]—would be the natural remedy. Whether the block-gradient long-run covariance estimator survives such a compression is open.

Geometric mixing is a real restriction, and the block length has to respect it. Theorem 10 buys $b \asymp \log N$ from the geometric decay in Lemma 12. Under algebraic mixing the flat-top bias is polynomial rather than exponential and the admissible b grows accordingly; long-memory streams are outside the theory altogether. This is not merely formal: our air-quality stream has a residual lag-one autocorrelation of 0.91 and a median variance-inflation factor of 9.9, and there the block length needed for calibrated intervals is an order of magnitude larger than for the traffic stream (Table 4). The free diagnostic $r_j = |u_j^\top(\widehat{S}^{\mathrm{FT}} - \widehat{S}^{\mathrm{BM}})u_j|/(u_j^\top \widehat{S}^{\mathrm{FT}} u_j)$ detects the problem—it estimates the relative block-length bias of plain batch means—but it does not remove it. It could be turned into an automatic rule by maintaining covariance accumulators at several block scales $b, 2b, 4b, \ldots$ at once (one extra $O(d^2)$ update per block per scale, all from the same block gradients) and reporting from the smallest scale at which r_j falls below a tolerance; we did not implement this, and we do not claim an automatic rule for b.

$\widehat{S}^{\mathrm{FT}}$ need not be positive semidefinite. This is intrinsic to flat-top and lugsail estimators [85, Remark 5]. Marginal intervals need only positive scalars $u_j^\top \widehat{S}^{\mathrm{FT}} u_j$. The documented fallback to $\widehat{S}^{\mathrm{BM}}$ never triggered in any simulation, but it does trigger on the real streams — at a rate of 0.0% of coordinate-replicate pairs — which is what one should expect, since there the two designs are ill-conditioned and the lag-one correction is comparable to the batch-means term. We report the rate rather than claiming it away, and a user who needs the full matrix (for a joint test, say) must project it.

The curvature form is required. Everything rests on $\nabla^2 F(\theta) = \mathbb{E}[\alpha\, \Psi\Psi^\top]$, which holds for generalised linear models, ridge and softmax regression, the geometric median and Gauss–Newton problems, but not for a general objective. Without it there is no rank-one recursion and no inversion-free update.

Warm-up is a real requirement, not an implementation detail. The $1/t$ schedule with a poor preconditioner leaves a slowly decaying component in the error. On the autoregressive design the effect is now mild — with a negligible warm-up, coverage of nominal 95% intervals is 0.928 rather than 0.949 — because the step-length safeguard absorbs most of what a bad preconditioner does. On the regime-switching design it is not mild: the relative variance is 2.19 at $c_{\mathrm{w}} = 50$ against 1.41 at $c_{\mathrm{w}} = 200$ (Table 5(c′)). We give a scaling rule ($c_{\mathrm{w}} d$ curvature updates, with $c_{\mathrm{w}} = 200$ throughout) and show insensitivity above it, but this is a tuning constant, and the asymptotic theory is silent about it. What sets its size is not the number of observations but the number of effectively distinct design points they span: on a persistent regime-switching stream, $50d$ consecutive observations cover only about thirty regime visits, and the curvature estimate is still rank-starved (Table 5(c′)). The warm-up also has a cost consequence we state rather than bury: it performs $c_{\mathrm{w}} d$ rank-one updates at $O(d^2)$ each, so the total time is $O(dN + c_{\mathrm{w}} d^3)$ and the advertised $O(dN)$ only bites once $N \gtrsim c_{\mathrm{w}} d^2$. With $c_{\mathrm{w}} = 200$ and $d = 40$ that means $N \gtrsim 3 \times 10^5$, so at the $N = 2 \times 10^5$ of Table 6 the warm-up dominates the streaming pass for every $d \geq 40$ we report, which is why that table separates the streaming count from the total.

Blocking creates a stability condition, and our fix is not fully proved. The contraction condition $\gamma_t \|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}} < 2$ is a consequence of blocking the gradient, so it is a cost of our construction and not of the method we build on. The step-length safeguard in (6) removes the failure it causes, and Proposition 1 shows that on the event where the safeguard is eventually inactive it changes no asymptotic statement. What we do *not* prove is that this event has probability one: the implication we establish runs from convergence to eventual inactivity, not back, because the safeguard factor depends on the current block's gradient and so is not predictable, which breaks the martingale structure the proofs use. Appendix C says exactly what would close the gap — an expanding-truncation construction [4, 52], or a predictable variant that caps using the previous block's amplification at an extra $O(d^2)$ per block — and we did neither. Instead we report the activation counts, which are between 2 and 7 per run and flat in N across every design and sample size in Tables 2 and 5. Two further caveats. The condition itself comes from the linearised recursion (7), so it is a heuristic for non-quadratic objectives, exact only in the linear-model case. And c_D is a tuning constant with a genuine failure mode: Table 5(i) shows it is irrelevant over a factor of sixteen on the designs that do not need the safeguard, but on the one that does, $c_D = 4$ is too loose to prevent the blow-up. The asymptotic theory says nothing about it, and we have no rule for choosing it beyond the value that worked across our designs.

The central limit theorem is proved for fixed b, and we recommend growing b. Theorem 6 is unconditional at each fixed block length, but Theorem 10 takes $b \asymp \log N$, so the interval we recommend sits in the triangular-array regime where our proof needs an extra hypothesis: that the finite random constants in Lemma 14 are bounded in probability uniformly in N along the sequence b_N. We state that hypothesis in Theorem 6 rather than hiding it, and we do not verify it. Closing the gap means redoing Lemma 14 with constants explicit in b; the naive route — bounding $\mathbb{E}\|H^{-1} - P_{t-1}\|_{\mathrm{op}}\|\varepsilon_t\|$ directly — is the one the appendix shows loses a factor $T^{1/2}$, which is why Gordin's decomposition is there in the first place. Two things make us believe the composition is sound in practice rather than merely convenient: growing b makes the block scores less dependent, not more, so the mechanism the constants come from is working in our favour; and coverage is flat in b over $b \in [10, 320]$ at $N = 200{,}000$ (Table 5(b)), which is the statement the two theorems need to compose.

Rates and what we do not prove. We give an almost sure rate and a central limit theorem, not a Berry–Esseen bound; the first-order Markovian literature has the latter for linear stochastic approximation [83], and we do not match it. We attain the semiparametric efficiency bound of Li et al. [57] rather than establishing it. Our $\widetilde{O}(N^{-1/2})$ rate for $\widehat{S}^{\mathrm{FT}}$ is not comparable to the minimax rate $\Theta(N^{-(1-a)/2})$ for Hessian-free covariance estimation from an iterate path [69], because we use more information (block gradients and a maintained curvature estimate) and because our step exponent is $a = 1$, outside the range that bound is stated for. Within the Hessian-free class the bound is close to attained [94], so the gap we report is about the information used, not about slack in their analysis. We make no claim of optimality within our own information set: we do not know the minimax rate for estimators that may use block gradients and a running curvature estimate.

Scope of the empirical evidence. All designs are convex M-estimation with $d \leq 160$ and $N \leq 4 \times 10^5$; there is no deep-learning experiment and the theory would not support one. The dependence in the simulated designs is stationary and geometrically decaying by construction. On the real streams we do not know the truth, which is why we run two protocols; the semi-synthetic protocol has an exact target but a synthetic (autoregressive) error process, and the fully real protocol has a real error process but a target that is itself estimated—we correct the nominal level for that exactly, but the correction assumes the segments are approximately exchangeable, which non-stationarity would violate. Neither protocol alone would be convincing; together they bracket the honest answer.

One thing we deliberately do not claim. Gapping improves the curvature estimate at matched cost (Proposition 11, Figure 4), but the curvature estimate enters the limit distribution only through a rate, so the end-to-end effect on the point estimate and on coverage is second order and we could not measure it reliably. The case for gapping is that it is free, deterministic, and provably better than the randomised alternative—not that it changes the headline numbers.

8 Conclusions

On a dependent stream, a streaming quasi-Newton method can get the estimate right and the uncertainty wrong. The error is not small and does not go away: the reported variance omits the autocovariances of the score, so a nominal 95% interval converges to a coverage below 95% and stays there. Proposition 8 says exactly where: at $2\Phi(z_{\alpha/2}/\sqrt{\kappa}) - 1$, a number computable from one variance-inflation factor κ. At the inflation we measure on hourly air quality this is 0.466, so more than half of the intervals a practitioner would report on that stream miss the value they are meant to cover.

The remedy is a change of reading order rather than an extra estimation stage. Consume the stream in contiguous blocks and take one preconditioned step per block. The block-averaged gradients are then, up to a factor, the non-overlapping batch means of the score process, and batch means are a standard estimator of exactly the long-run variance the interval needs. So the object the statistics requires and the object the computation already produces are the same, and the resulting interval is *cheaper* than the invalid one: an instantaneous plug-in covariance costs $O(d^2)$ per observation, while the blocked long-run covariance costs $O(d^2)$ per block. Gapping the rank-one curvature updates is the second change, and it is a cost knob rather than an inference device — it dominates the randomised thinning the independent-data method uses, at the same number of updates, and makes the per-block cost deterministic.

Three practical consequences are worth stating in plain terms. First, block length is the one parameter that needs thought, and it needs less than one might fear: because the blocks carry gradients rather than parameter values, their length is set by how quickly the *data* forget, not by how quickly the optimiser settles, so b of order $\log N$ suffices where iterate-based methods need a power of N. A diagnostic that costs nothing, r_j, indicates when b is too short. Second, blocking has a price we pay and report: it makes a step-size condition non-trivial, and without the $O(d)$ safeguard of (6) our own method diverges on a persistent regime-switching design. Anyone who blocks a preconditioned step on dependent data will meet this, and Section 2 explains why and what to do. Third, honest intervals are wider. On our designs they are 1.37–2.72 times the dependence-blind width, which is $\sqrt{\kappa}$ and not a tuning choice. The narrower interval was never correct; it was the wrong length.

What we have not done bounds the claims. The central limit theorem is proved unconditionally for each fixed block length, while the estimator we recommend lets b grow with N; the safeguard is shown to be asymptotically invisible on the event that it stops acting, not unconditionally; memory remains $O(d^2)$, so very high-dimensional streams need a sketched curvature representation whose interaction with block-gradient variance estimation is open; and geometric mixing excludes long-memory data. Section 7 states each of these precisely. The construction itself is general: any streaming method that forms block gradients and maintains a curvature estimate can report a long-run variance at no extra order of cost, and we would expect the same argument to carry to sketched and diagonal curvature approximations.

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A Proofs

Throughout, C denotes a finite constant that may change from line to line, $\delta_t = \theta_t - \theta^\star$, $\varepsilon_t = \bar{g}_t(\theta^\star)$, $P_t = \bar{H}_t^{-1}$, and $\Lambda_k = \mathbb{E}[\varepsilon_0 \varepsilon_k^\top]$. We take $\gamma_t = 1/t$ throughout and ignore the step-length safeguard Π_t of (6); Proposition 1, proved in Appendix C, shows that on the event where Π_t is eventually inactive the safeguarded iterates coincide with these from a finite time on, and Appendix C.1 treats the rejected step-size-shift variant $\gamma_t = (t + t_0)^{-1}$ separately. We write n_t for the number of rank-one curvature updates completed by the end of block t; with gap m and no Bernoulli thinning, $n_t = \lfloor tb/m \rfloor$, so $n_t \asymp t$.

Order of the argument. The results are proved in the order of Remark 2: Lemma 2 (from the ridge alone, no assumption on the iterates); Theorem 4 (uses Lemma 2 only); the preliminary rate $\|\delta_t\| = O(t^{-1/2} \log^c t)$ (uses Theorem 4 and consistency of $\bar{H}$, not its rate); Theorem 3; and then Theorems 5, 6, 9 and 10. Every remainder bound below is uniform over $b \le C \log N$, which is the only regime in which Theorem 10 uses them, and the b-dependence is displayed wherever it is not $O(1)$.

A.1 Preliminaries on dependence

Lemma 12 (Geometric decay of the score autocovariances)**.** *Under Assumption 5, $\|\Gamma_k\|_{\mathrm{op}} \le C\rho^{|k|}$ with $\rho = c_{\mathrm{mix}}^{\nu/(2+\nu)}$, and $\|\Lambda_k\|_{\mathrm{op}} \le Cb^{-1}\rho^{(|k|-1)b}$ for $|k| \ge 1$ while $\|\Lambda_0\|_{\mathrm{op}} \le C/b$. Consequently $\sum_{k\in\mathbb{Z}} \|\Lambda_k\|_{\mathrm{op}} \le C/b$ and $S = \sum_k \Gamma_k$ converges absolutely.*

Proof. Davydov's inequality [18], in the form of Rio [76, Cor. 1.1], gives for random variables $U \in \sigma(\xi_i, i \le 0)$, $V \in \sigma(\xi_i, i \ge k)$ with $2+\nu$ moments, $|\operatorname{Cov}(U, V)| \le 8\,\alpha_{\mathrm{mix}}(k)^{\nu/(2+\nu)} \|U\|_{2+\nu} \|V\|_{2+\nu}$. Applying it coordinatewise to g_0 and g_k and using Assumption 5 gives the first claim. For the second, $\Lambda_k = b^{-2} \sum_{i\in B_0} \sum_{j \in B_k} \Gamma_{j-i}$; for $k \ge 1$ every lag $j-i$ is at least $(k-1)b+1$, so $\|\Lambda_k\|_{\mathrm{op}} \le b^{-2} \cdot b^2 \cdot C\rho^{(k-1)b}$, and for $k = 0$, $\|\Lambda_0\|_{\mathrm{op}} \le b^{-2} \sum_{|l|<b} (b - |l|) C \rho^{|l|} \le Cb^{-1}$. Summing the geometric series gives $\sum_k \|\Lambda_k\|_{\mathrm{op}} \le C/b$. ◻

Lemma 13 (Martingale–coboundary decomposition)**.** *Let $(\zeta_t)_{t\ge1}$ be stationary, centred, $\mathbb{R}^q$-valued, $\mathcal{F}_t$-measurable with $\mathcal{F}_t=\sigma(\xi_i : i\le tb)$, and let $\mathbb{E}\|\zeta_1\|^{2+\nu}<\infty$ under Assumption 5. Then $\sum_{n\ge0}\|\mathbb{E}[\zeta_n\mid\mathcal{F}_0]\|_2<\infty$, the series $z_t=\sum_{n\ge0}\mathbb{E}[\zeta_{t+n}\mid\mathcal{F}_t]$ converges in L^2, and*

$$\zeta_t \;=\; m_{t+1}+z_t-z_{t+1}, \qquad m_{t+1}=z_{t+1}-\mathbb{E}[z_{t+1}\mid\mathcal{F}_t], \tag{14}$$

where (m_t) is a stationary martingale difference sequence with respect to $(\mathcal{F}_t)$ and $\mathbb{E}\|m_1\|^2<\infty$, $\mathbb{E}\|z_1\|^2<\infty$. Moreover $\mathbb{E}[m_1m_1^\top]=b^{-1}S$ when $\zeta_t=\varepsilon_t$.

Proof. For U that is $\sigma(\xi_i : i\ge n)$-measurable with $2+\nu$ moments, the standard consequence of strong mixing (see Rio 77, Ch. 1) is $\|\mathbb{E}[U\mid\mathcal{F}_0]\|_2\le C\,\alpha_{\mathrm{mix}}(n)^{\nu/(2(2+\nu))}\|U\|_{2+\nu}$, which under Assumption 5 decays geometrically in n and is therefore summable; this is Gordin's condition, and (14) together with the stated moment bounds is Gordin's decomposition; see Hall and Heyde [37, Ch. 5]. Summing (14) telescopes the coboundary, so $\sum_{t\le T}\zeta_t=\sum_{t\le T}m_{t+1}+z_1-z_{T+1}$; dividing by $\sqrt{T}$ and using $z_{T+1}/\sqrt{T}\to0$ in probability identifies $\lim T^{-1}\operatorname{Var}(\sum_{t\le T}\zeta_t)=\mathbb{E}[m_1m_1^\top]$, which for $\zeta_t=\varepsilon_t$ is $b^{-1}S$ by (1). ◻

Lemma 14 (Weighted sums with adapted weights: convergence and rate)**.** *Let (ζ_t) be as in Lemma 13, let (A_t) be $\mathcal{F}_{t-1}$-measurable random matrices, and let (γ_t) be deterministic and non-increasing. Suppose there are deterministic (c_t), (u_t) and a finite random Λ with, almost surely for all t,*

$$\|A_t\|_{\mathrm{op}}\le\Lambda c_t \qquad \text{and} \qquad \|\gamma_{t+1}A_{t+1}-\gamma_tA_t\|_{\mathrm{op}}\le\Lambda u_t, \tag{15}$$

and write $V_T=\sum_{t\le T}\gamma_t^2c_t^2$ and $U_T=\sum_{t\le T}u_t$. Then for every $\varpi>0$, almost surely,

$$\Big\|\sum_{t\le T}\gamma_tA_t\zeta_t\Big\|=O\big(V_T^{1/2}\log^{1/2+\varpi}T\big)+O\big(U_T\log^{1+\varpi}T\big)+O\big(\gamma_Tc_TT^{1/2+\varpi}\big). \tag{16}$$

If moreover $V_\infty<\infty$, $U_\infty<\infty$ and $\gamma_Tc_TT^{1/2+\varpi}\to0$ for some $\varpi>0$, then $\sum_t\gamma_tA_t\zeta_t$ converges almost surely.

Proof. Localisation. For $K\in\mathbb{N}$ replace A_t by $A_t\mathbf{1}\{\sup_{s\le t}\max(\|A_s\|_{\mathrm{op}}/c_s,\ \|\gamma_{s+1}A_{s+1}-\gamma_sA_s\|_{\mathrm{op}}/u_s)\le K\}$, which is still $\mathcal{F}_{t-1}$-measurable (the indicator is decreasing in t and $\mathcal{F}_{t-1}$-measurable at each t because it involves A_s only for $s\le t$, and A_{s+1} enters only through $s+1\le t$). On $\{\Lambda\le K\}$ the two processes agree and $\mathbb{P}(\bigcup_K\{\Lambda\le K\})=1$, so it suffices to prove the claims under the *deterministic* envelopes $\|A_t\|_{\mathrm{op}}\le Kc_t$, $\|\gamma_{t+1}A_{t+1}-\gamma_tA_t\|_{\mathrm{op}}\le Ku_t$. Fix K and drop it.

Martingale part. Insert (14): $M_T=\sum_{t\le T}\gamma_tA_tm_{t+1}$ is a martingale, because A_t is $\mathcal{F}_t$-measurable and $\mathbb{E}[m_{t+1}\mid\mathcal{F}_t]=0$ — this is the exact cancellation the decomposition buys, and it holds whatever A_t does. Its predictable quadratic variation obeys

$$\langle M\rangle_T=\sum_{t\le T}\gamma_t^2\,\mathbb{E}\big[\|A_tm_{t+1}\|^2\bigm|\mathcal{F}_t\big]\le K^2\sum_{t\le T}\gamma_t^2c_t^2\,v_t, \qquad v_t:=\mathbb{E}\big[\|m_{t+1}\|^2\mid\mathcal{F}_t\big],$$

with (v_t) stationary, non-negative and $\mathbb{E}v_t=\mathbb{E}\|m_1\|^2<\infty$. By Tonelli $\mathbb{E}\big[\sum_{t\le T}\gamma_t^2c_t^2v_t\big]=\mathbb{E}\|m_1\|^2V_T$, so Markov's inequality along $T=2^j$ and Borel–Cantelli give $\langle M\rangle_T=O(V_T\log^{\varpi}T)$ a.s. (and $\langle M\rangle_\infty<\infty$ a.s. if $V_\infty<\infty$). The martingale law of the iterated logarithm then gives $\|M_T\|=O(V_T^{1/2}\log^{1/2+\varpi}T)$ a.s., and if $\langle M\rangle_\infty<\infty$ a locally square-integrable martingale converges a.s.

Coboundary part. Abel summation gives

$$\sum_{t\le T}\gamma_tA_t(z_t-z_{t+1})=\gamma_1A_1z_1-\gamma_TA_Tz_{T+1}+\sum_{t<T}\big(\gamma_{t+1}A_{t+1}-\gamma_tA_t\big)z_{t+1}.$$

The last sum is at most $K\sum_{t<T} u_t\|z_{t+1}\|$, a non-negative sum with expectation $K\mathbb{E}\|z_1\|U_T$, hence $O(U_T\log^{1+\varpi}T)$ a.s. by Markov and Borel–Cantelli along dyadic blocks; if $U_\infty<\infty$ its expectation is finite and it converges a.s. Finally $\|\gamma_T A_T z_{T+1}\|\le K\gamma_T c_T\|z_{T+1}\| = O(\gamma_T c_T T^{1/2+\varpi})$ a.s., because $\mathbb{E}\|z_1\|^2<\infty$ and Borel–Cantelli give $\|z_T\| = o(T^{1/2+\varpi})$. ◻

Remark 5 (Why the lemma is stated this way)**.** Three features are load-bearing and each corresponds to a place where a naive argument fails. (i) The weights are *random and adapted*: they contain $\bar H_{t-1}^{-1}$, so a lemma for deterministic weights would not apply. Note that adaptedness is essential and not cosmetic: for bounded adapted weights and a mixing, centred, stationary ζ the sum $\sum_t a_t\zeta_t$ can grow like T if a_t is allowed to correlate with ζ_t — take $\zeta_t = u_t + u_{t-1}$ with u_t i.i.d. signs and $a_t = \zeta_{t-1}$ — which is exactly the non-martingale obstruction. The decomposition (14) removes it because $\mathbb{E}[m_{t+1}\mid\mathcal{F}_t]=0$ *exactly*, whatever A_t does. (ii) The envelope is deterministic only up to a finite random factor, which is all that is available for weights built from the iterates; the localisation handles it. (iii) We use a martingale law of the iterated logarithm rather than a Bernstein inequality for mixing sums. The Bernstein inequality of Merlevède et al. [62] is stated for uniformly *bounded* summands, which Assumption 5 does not provide; after (14) the summands form a martingale difference sequence and no such inequality is needed. (iv) The three terms of (16) are needed separately: the applications below use $\gamma_t\equiv 1$, for which U_T can diverge logarithmically, so a hypothesis $U_\infty<\infty$ would be too strong.

Lemmas 13 and 14 are the technical substitute for the martingale argument used in the independent-data analysis. Two features matter. First, the weights may be *random and adapted*, which is what we need because they involve $\bar H_{t-1}^{-1}$. Second, the conditions (15) are of the form already familiar from the independent-data theory. The vanishing ridge supplies the deterministic envelope: it forces $\lambda_{\min}(A_n)\ge c\,n^{1-q_r}$, hence $\|\bar H_t^{-1}\|_{\rm op} = W_{n_t}\|A_{n_t}^{-1}\|_{\rm op}\le c^{-1}n_t^{q_r} =: c_t$ with $q_r<1/2$, so with $\gamma_t = 1/t$ the first half of (15) is $\sum_t O(t^{2q_r-2})<\infty$. The second half holds because a single rank-one Riccati update changes the preconditioner by only $O(1/t)$:

Lemma 15 (Increments of the preconditioner)**.** (a) From the ridge alone. *Under Assumption 3 and the ridge of* (4) *— so using only Lemma 2, with no assumption on the iterates —* $\|\bar H_t^{-1}-\bar H_{t-1}^{-1}\|_{\rm op} = O(t^{2q_r-1})$ *a.s.; hence with $\gamma_t = 1/t$ and $A_t = \bar H_{t-1}^{-1}$, $\|\gamma_{t+1}A_{t+1}-\gamma_t A_t\|_{\rm op} = O(t^{2q_r-2})$ a.s., which is summable for $q_r<1/2$, so the second half of* (15) *holds with $U_\infty<\infty$.*

(b) Sharper, once the curvature has converged. *Under the assumptions of Theorem 3,* $\|\bar H_t^{-1}-\bar H_{t-1}^{-1}\|_{\rm op} = O(1/t)$ *a.s., so with $\gamma_t\equiv 1$ and $A_t = \bar H_{t-1}^{-1}$ the increments are $u_t = O(1/t)$ and $U_T = O(\log T)$.*

Proof. $\bar H_t^{-1} = W_{n_t}A_{n_t}^{-1}$, and from block $t-1$ to block t the weight W increases by $O(1)$ while A^{-1} changes by $O(1)$ rank-one Sherman–Morrison steps $A^{-1}\mapsto A^{-1} - cA^{-1}uu^\top A^{-1}/(1+cu^\top A^{-1}u)$. Writing $v = A^{-1}u$, the operator norm of one such step is $c\|v\|^2/(1+cu^\top v)$, and since $u^\top v = v^\top Av\ge\lambda_{\min}(A)\|v\|^2$ we get the two bounds

$$\|\Delta A^{-1}\|_{\rm op}\ \le\ \min\Big\{\frac{1}{\lambda_{\min}(A)},\ \frac{c\|u\|^2}{\lambda_{\min}(A)^2}\Big\}. \tag{17}$$

For (a), Lemma 2 gives $\lambda_{\min}(A_n)\ge c'n^{1-q_r}$, so the second bound in (17) is $O(n^{2q_r-2})$; multiplying by $W_{n_t} = O(t)$ and using $n_t\asymp t$ gives $\|\bar H_t^{-1}-\bar H_{t-1}^{-1}\|_{\rm op} = O(t^{2q_r-1})$, and the change in W contributes $O(1)\cdot\|A^{-1}\|_{\rm op} = O(t^{q_r-1})$, which is smaller. The stated bound on $\gamma_{t+1}A_{t+1}-\gamma_t A_t$ then follows from $\gamma_t = 1/t$ together with $\gamma_{t+1}-\gamma_t = O(t^{-2})$ and $\|A_t\|_{\rm op} = O(t^{q_r})$. For (b), Theorem 3 gives $A_{n_t}/n_t\to H\succ 0$, so $\lambda_{\min}(A_{n_t})\ge c't$ a.s. for large t, the second bound in (17) is $O(t^{-2})$, and multiplying by $W_{n_t} = O(t)$ gives $O(1/t)$. ◻

A.2 The ridge bound (Lemma 2)

Each curvature update adds $\iota_n e_{k_n} e_{k_n}^\top \succeq 0$ along a cycling basis direction, and the data term $\alpha_{i_n} \Psi_{i_n} \Psi_{i_n}^\top$ is positive semidefinite, so it can only help. After n updates every basis direction has received at least $\lfloor n/d \rfloor$ ridge increments, and dropping the data terms,

$$A_n \;\succeq\; \lambda_0 I + \Big(\tfrac{1}{d} \textstyle\sum_{k \le n} \iota_k\Big) I. \tag{18}$$

Now the point of freezing the ridge scale. Because $\iota_k = c_\iota \varsigma_\star k^{-q_\mathrm{r}}$ with $\varsigma_\star$ a *constant* — measurable with respect to the first curvature-bearing block and, by Assumption 3, almost surely finite and strictly positive whenever that block's curvature is not identically zero — we may sum:

$$\sum_{k \le n} \iota_k = c_\iota \varsigma_\star \sum_{k \le n} k^{-q_\mathrm{r}} \;\ge\; \frac{c_\iota \varsigma_\star}{1 - q_\mathrm{r}} \left(n^{1-q_\mathrm{r}} - 1\right),$$

using $\sum_{k \le n} k^{-q_\mathrm{r}} \ge \int_1^{n+1} x^{-q_\mathrm{r}} dx$ and $q_\mathrm{r} < 1$. Hence $\lambda_{\min}(A_n) \ge c' n^{1-q_\mathrm{r}}$ for all $n \ge 2$ with $c' = c_\iota \varsigma_\star / \{2d(1 - q_\mathrm{r})\}$, and since $W_{n_t} = 1 + n_t$,

$$\|\bar{H}_t^{-1}\|_{\mathrm{op}} = W_{n_t} \|A_{n_t}^{-1}\|_{\mathrm{op}} \le \frac{1 + n_t}{c' n_t^{1-q_\mathrm{r}}} = O(n_t^{q_\mathrm{r}}), \qquad \lambda_{\min}(\bar{H}_t) \ge \frac{c' n_t^{1-q_\mathrm{r}}}{1 + n_t} \;\ge\; c_1 n_t^{-q_\mathrm{r}}$$

with $c_1 = c'/2$ for $n_t \ge 1$. Nothing about the iterates is used, and no lower bound on the accumulated data terms is needed — which is exactly why the lemma may sit at the bottom of the ladder in Remark 2.

Remark 6 (Why the scale must be frozen)**.** Had we scaled the ridge by the *running* harmonic mean $d/\mathrm{tr}(\bar{H}_k^{-1})$ — the natural choice, since it is available for free from the maintained inverse — this proof would fail, and not for a technical reason. That scale is bounded above by $d\lambda_{\min}(\bar{H}_k)$, so a path along which the accumulated data terms contribute negligibly in some direction would have its ridge shrink in step with the very eigenvalue the ridge is supposed to hold up. Concretely, the only unconditional bound then available is $\varsigma_k \ge \lambda_0 / W_k = \Theta(1/k)$, giving $\sum_{k \le n} \iota_k = O(1)$ instead of $\Omega(n^{1-q_\mathrm{r}})$, and a Gronwall argument in the opposite direction shows this is tight: $\lambda_{\min}(A_n)$ can stay $O(1)$, i.e. $\|\bar{H}_n^{-1}\|_{\mathrm{op}} = \Omega(n)$, which would break the square-summability $\sum_t \gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 < \infty$ that Theorem 4 needs. The bound does hold once $\bar{H}_t \to H \succ 0$, but that is Theorem 3, three rungs higher, so the argument would be circular. Freezing the scale removes the problem at no cost: $\varsigma_\star$ serves only to make c_ι dimensionless, and Table 5(d) shows the results are unchanged if the scale is dropped altogether.

A.3 Consistency (Theorem 4)

Substituting $\bar{g}_t(\theta_{t-1}) = \nabla F(\theta_{t-1}) + \zeta_t$ into (6), where $\zeta_t = \bar{g}_t(\theta_{t-1}) - \nabla F(\theta_{t-1})$, and expanding $F(\theta_t) - F(\theta^\star)$ to second order using Assumption 1,

$$\begin{aligned} F(\theta_t) - F(\theta^\star) \le F(\theta_{t-1}) - F(\theta^\star) - \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \nabla F(\theta_{t-1}) \\ - \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \zeta_t + \tfrac{L}{2} \gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 \|\bar{g}_t(\theta_{t-1})\|^2. \end{aligned} \tag{19}$$

The last term is summable: $\gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 = O(t^{2q_\mathrm{r}-2})$ by Lemma 2 with $q_\mathrm{r} < 1/2$, and $\mathbb{E}\|\bar{g}_t(\theta_{t-1})\|^2 \le C + C'(F(\theta_{t-1}) - F(\theta^\star))$ by Assumption 2 and Jensen. The cross term is the one that would be handled by a martingale argument under independence. Instead, decompose $\zeta_t = \varepsilon_t + r_t$ with $r_t = \bar{g}_t(\theta_{t-1}) - \nabla F(\theta_{t-1}) - \varepsilon_t$, so that

$$\|r_t\| \le \|\widehat{H}_t - H\|_{\mathrm{op}} \|\delta_{t-1}\| + L_\delta \|\delta_{t-1}\|^2 \tag{20}$$

where $\widehat{H}_t$ is the block curvature; r_t is dominated by $\|\delta_{t-1}\|$ times an L^2-bounded factor, and its contribution to (19) is absorbed by the Robbins–Siegmund argument [79]. The remaining term $-\sum_t \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1}\varepsilon_t$ is handled by Lemma 14 with $A_t = \nabla F(\theta_{t-1})^\top P_{t-1}$, which is $\mathcal{F}_{t-1}$-measurable as required. Its first condition holds because $\|P_{t-1}\|_{\mathrm{op}} = O(t^{q_{\mathrm{r}}})$ with $q_{\mathrm{r}} < 1/2$ (Lemma 2) and $\mathbb{E}\|\nabla F(\theta_{t-1})\|^2$ is bounded on the events $\{\sup_{s\le t}\|\theta_s\| \le K\}$, over which we localise and then let $K \to \infty$; its second condition holds by Lemma 15(a) — which, being a consequence of Lemma 2 alone, keeps this rung free of Theorem 3 — together with $\|\nabla F(\theta_t) - \nabla F(\theta_{t-1})\|_{\mathrm{op}} \le L\|\theta_t - \theta_{t-1}\| = O(\gamma_t t^{q_{\mathrm{r}}})$. Robbins–Siegmund then gives $F(\theta_t) \to F(\theta^\star)$ a.s. and $\sum_t \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1}\nabla F(\theta_{t-1}) < \infty$ a.s.; and since $\gamma_t \lambda_{\min}(P_{t-1}) \ge \gamma_t/\|\bar{H}_{t-1}\|_{\mathrm{op}} \ge c\, t^{-1}$ (because $\|\bar{H}_t\|_{\mathrm{op}}$ is bounded above a.s., $\bar{H}_t$ being an average of terms with $\eta' > 2$ moments) we have $\sum_t \gamma_t \lambda_{\min}(P_{t-1}) = \infty$; Assumption 4 then forces $\theta_t \to \theta^\star$ a.s. ◻

A.4 The preliminary rate (rung (iii) of Remark 2)

This rung is what Theorem 3 needs and Theorem 4 does not give: a rate, not just convergence. We prove it using only Lemma 2 and Theorem 4.

Lemma 16 (Preliminary rate)**.** *Under Assumptions 1 to 5, with $q_{\mathrm{r}} \in (0, 1/4)$, $\|\theta_T - \theta^\star\| = O(T^{-1/2}\log^{1+\varpi} T)$ almost surely, for every $\varpi > 0$.*

Proof. Abel summation of (6) exactly as in (23) — an algebraic identity that uses only $\gamma_t = 1/t$ — gives (24): $\delta_T = -T^{-1}H^{-1}\sum_{t\le T}\varepsilon_t + (\mathrm{A}) + (\mathrm{B}) - (\mathrm{C})$. We bound the three remainders using only what rungs (i) and (ii) provide, namely $\|P_{t-1}\|_{\mathrm{op}} = O(t^{q_{\mathrm{r}}})$ a.s. (Lemma 2), $\theta_t \to \theta^\star$ a.s. (Theorem 4), and hence, by the local Lipschitz property in Assumption 3 and the continuity in Assumption 2, $\|P_{t-1} - H^{-1}\|_{\mathrm{op}} = o(1)$ and $\|I - P_{t-1}H\|_{\mathrm{op}} = o(1)$ a.s. Write $\rho_t = \|I - P_{t-1}H\|_{\mathrm{op}}$, so $\rho_t \to 0$ a.s.

Leading term. $T^{-1}\sum_{t\le T}\varepsilon_t = O(T^{-1/2}\log^{1/2+\varpi} T)$ a.s. by the martingale law of the iterated logarithm applied to (14) (the coboundary telescopes to $O(T^{-1})$ after division by T, using $\mathbb{E}\|z_1\|^2 < \infty$ and Borel–Cantelli along $\|z_t\| = o(t^{1/2})$ a.s.).

Term (A). Apply (16) with $\gamma_t \equiv 1$ and $A_t = H^{-1} - P_{t-1}$: the envelope may be taken $c_t \equiv 1$ (a finite random constant, since $\|P_{t-1}\|_{\mathrm{op}} = O(t^{q_{\mathrm{r}}})$ makes $\sup_t\|A_t\|_{\mathrm{op}}/c_t$ finite only after we note $A_t \to 0$; formally take $c_t = 1$ and $\Lambda = \sup_t\|A_t\|_{\mathrm{op}} < \infty$ a.s., which holds because $A_t \to 0$ and each A_t is finite). By Lemma 15(a) the increments are $u_t = O(t^{2q_{\mathrm{r}}-1})$, so $V_T = O(T)$ and $U_T = O(T^{2q_{\mathrm{r}}})$, and (16) gives $\|\sum_{t\le T} A_t\varepsilon_t\| = O(T^{1/2}\log^{1/2+\varpi} T) + O(T^{2q_{\mathrm{r}}}\log^{1+\varpi} T) + O(T^{1/2+\varpi})$. Dividing by T and using $q_{\mathrm{r}} < 1/4$, which is exactly where that restriction is needed, $(\mathrm{A}) = O(T^{-1/2}\log^{1/2+\varpi} T) + O(T^{2q_{\mathrm{r}}-1}\log^{1+\varpi} T) = O(T^{-1/2}\log^{1/2+\varpi} T)$ a.s.

Terms (B) and (C): the bootstrap. $(\mathrm{B}) = T^{-1}\sum_{t\le T}(I - P_{t-1}H)\delta_{t-1}$ so $\|(\mathrm{B})\| \le \big(\sup_{t\le T}\rho_t\big)\,\bar{\delta}_T$ with $\bar{\delta}_T = T^{-1}\sum_{t\le T}\|\delta_{t-1}\|$, and by (20) $\|(\mathrm{C})\| \le CT^{-1}\sum_{t\le T}\|P_{t-1}\|_{\mathrm{op}}(\|\widehat{H}_t - H\|_{\mathrm{op}}\|\delta_{t-1}\| + \|\delta_{t-1}\|^2)$, which is $o(1)\,\bar{\delta}_T$ a.s. because $\|P_{t-1}\|_{\mathrm{op}}\|\widehat{H}_t - H\|_{\mathrm{op}}$ and $\|P_{t-1}\|_{\mathrm{op}}\|\delta_{t-1}\|$ are both $o(1)$ a.s. ($\delta_t \to 0$ by Theorem 4, and $\|P_{t-1}\|_{\mathrm{op}}\|\widehat{H}_t - H\|_{\mathrm{op}} \to 0$ because $\|P_{t-1}\|_{\mathrm{op}} \to \|H^{-1}\|_{\mathrm{op}}$ and $\widehat{H}_t \to H$ in the Cesàro sense along the block index). Hence there is a random $T_1 < \infty$ and a sequence $\epsilon_T \downarrow 0$ with

$$D_T := \max_{t\le T}\|\delta_t\| \;\le\; \epsilon_T D_T + R_T, \qquad R_T = O(T^{-1/2}\log^{1/2+\varpi} T) \text{ a.s.,} \tag{21}$$

for $T \ge T_1$, where we used $\bar{\delta}_T \le D_T$ and that the same bound applies to every $t \le T$ (the right-hand side is non-decreasing in T). For T large enough that $\epsilon_T \le 1/2$, (21) rearranges to $D_T \le 2R_T$, which is the claim with $\log^{1+\varpi}$ absorbing $\log^{1/2+\varpi}$. ◻

Two things are worth flagging about this rung. It is the only place the restriction $q_{\mathrm{r}} < 1/4$ (rather than $q_{\mathrm{r}} < 1/2$) is used, and it is used because at this rung the sharper increment bound Lemma 15(b) is not yet available. And the bootstrap in (21) is why the rate carries a log factor rather than being clean $T^{-1/2}$; Theorem 5 recovers the sharper statement once Theorem 3 is in hand.

A.5 The curvature rate (Theorem 3)

Write $Y_n = \alpha(\theta^\star;\xi_{i_n})\Psi(\theta^\star;\xi_{i_n})\Psi(\theta^\star;\xi_{i_n})^\top$ and $\tilde{Y}_n = \alpha(\theta_{t(n)-1};\xi_{i_n})\Psi\Psi^\top$ for the term actually accumulated, where $t(n)$ is the block containing i_n. Then, by (4) and (5),

$$\bar{H}_t - H = \underbrace{\frac{\lambda_0\varsigma_0 I}{W_{n_t}}}_{\text{(i)}} + \underbrace{\frac{1}{W_{n_t}}\sum_{n\le n_t}\iota_n e_{k_n}e_{k_n}^\top}_{\text{(ii)}} + \underbrace{\frac{1}{W_{n_t}}\sum_{n\le n_t}(Y_n - H)}_{\text{(iii)}} + \underbrace{\frac{1}{W_{n_t}}\sum_{n\le n_t}(\tilde{Y}_n - Y_n)}_{\text{(iv)}}. \tag{22}$$

Since $W_{n_t} = 1 + n_t \asymp t$, term (i) is $O(1/t)$. For (ii), $\|\cdot\|_{\mathrm{op}} \le n_t^{-1}\sum_{n\le n_t}\iota_n \le C n_t^{-q_{\mathrm{r}}}$ because $\iota_n \le C n^{-q_{\mathrm{r}}}$ and $q_{\mathrm{r}} < 1$; the same bound from below along each basis direction gives, together with $H \succ 0$, $\lambda_{\min}(\bar{H}_t) \ge c\, n_t^{-q_{\mathrm{r}}}$ for all t (this is the only place the ridge is used, and it is what bounds $\|\bar{H}_t^{-1}\|_{\mathrm{op}}$ for *every* t rather than eventually). Term (iv) is bounded, by the local Lipschitz property in Assumption 3 and Cauchy–Schwarz, by $Cn_t^{-1}\sum_{n\le n_t}\|\theta_{t(n)-1}-\theta^\star\|\cdot\|\Psi\Psi^\top\|_{\mathrm{op}}$, which by Lemma 16 is $O(n_t^{-1}\sum_{n\le n_t} n^{-1/2}\log^{1+\varpi} n) = O(n_t^{-1/2}\log^{1+\varpi} n_t)$ a.s.

The new content is term (iii). The array $(Y_n)_{n\ge1}$ is the subsequence of a stationary geometrically mixing sequence at spacing m, hence itself stationary with mixing coefficients $\alpha_{\mathrm{mix}}(jm) \le K c_{\mathrm{mix}}^{jm}$, and has $\eta' > 2$ moments by Assumption 3. Applying Lemma 14 with $a_n \equiv 1$ (i.e. its maximal-inequality half, which does not need square-summability) gives $\|\sum_{n\le n_t}(Y_n - H)\|_{\mathrm{op}} = O(n_t^{1/2}\log^{1/2+\varpi} n_t)$ a.s., so (iii) $= O(t^{-1/2}\log^{1/2+\varpi} t)$ a.s. Collecting the four terms, $\|\bar{H}_t - H\|_{\mathrm{op}} = O(t^{-\min\{q_{\mathrm{r}},1/2\}}\log^{c'} t)$ a.s., which is $O(t^{-\nu_H})$ for every $\nu_H < \min\{q_{\mathrm{r}}, 1/2\}$. Finally, on the event $\|\bar{H}_t - H\|_{\mathrm{op}} \le \lambda_{\min}(H)/2$ (which holds for all large t a.s.), $\|\bar{H}_t^{-1} - H^{-1}\|_{\mathrm{op}} \le \|\bar{H}_t^{-1}\|_{\mathrm{op}}\|\bar{H}_t - H\|_{\mathrm{op}}\|H^{-1}\|_{\mathrm{op}} \le 2\lambda_{\min}(H)^{-2}\|\bar{H}_t - H\|_{\mathrm{op}}$. □

A.6 Proof of Theorem 5 (rate) and Proposition 7 and Theorem 6 (CLT)

Write (6) as $\delta_t = \delta_{t-1} - \gamma_t P_{t-1}\{H\delta_{t-1} + \varepsilon_t + r_t\}$ with r_t as above. Multiplying by t and using $\gamma_t = 1/t$,

$$t\,\delta_t = (t-1)\delta_{t-1} + (I - P_{t-1}H)\delta_{t-1} - P_{t-1}\varepsilon_t - P_{t-1}r_t, \tag{23}$$

so that, summing from $t = 1$ to T (Abel summation) and dividing by T,

$$\delta_T = -\frac{1}{T}H^{-1}\sum_{t\le T}\varepsilon_t \;+\; \underbrace{\frac{1}{T}\sum_{t\le T}(H^{-1} - P_{t-1})\varepsilon_t}_{\text{(A)}} \;+\; \underbrace{\frac{1}{T}\sum_{t\le T}(I - P_{t-1}H)\delta_{t-1}}_{\text{(B)}} \;-\; \underbrace{\frac{1}{T}\sum_{t\le T}P_{t-1}r_t}_{\text{(C)}}. \tag{24}$$

The leading term is *exactly* $-N_T^{-1}H^{-1}\sum_{i\le N_T} g_i$, because $\sum_{t\le T}\varepsilon_t = b^{-1}\sum_{i\le N_T} g_i$ and $N_T = bT$. This identity is Proposition 7: the leading term does not depend on b.

Term (A). Apply (16) with $\gamma_t \equiv 1$, $\zeta_t = \varepsilon_t$ and the $\mathcal{F}_{t-1}$-measurable weights $A_t = H^{-1} - P_{t-1}$: by Theorem 3 the envelope is $c_t = t^{-\nu_H}$, and by Lemma 15(b) the increments are $u_t = O(t^{-1})$. Hence $V_T = O(T^{1-2\nu_H})$ (as $\nu_H < 1/2$), $U_T = O(\log T)$, and $\gamma_T c_T T^{1/2+\varpi} = O(T^{1/2-\nu_H+\varpi})$, so

$$\Big\|\sum_{t\le T}(H^{-1} - P_{t-1})\varepsilon_t\Big\| = O\big(T^{1/2-\nu_H}\log^{1/2+\varpi} T\big) + O\big(\log^{2+\varpi} T\big) + O\big(T^{1/2-\nu_H+\varpi}\big) = O\big(T^{1/2-\nu_H+\varpi}\big) \text{ a.s.}$$

Choosing $\varpi < \nu_H$ gives (A) $= O(T^{-1/2-\nu_H+\varpi}) = o(T^{-1/2})$ a.s. It is worth seeing what does the work. The only bound available without (14) is $\|\mathbb{E}[(H^{-1} - P_{t-1})\varepsilon_t]\| \le \|H^{-1} - P_{t-1}\|_{\mathrm{op}}\|\varepsilon_t\|_2 = O(t^{-\nu_H} b^{-1/2})$ — Davydov gives no decay across a one-block gap, since the weight and the summand are only one block apart — and averaging that over $t \le T$ leaves $O(b^{-1/2}T^{-\nu_H})$, which is *not* $o(T^{-1/2})$ for any admissible $\nu_H < 1/2$. The extra $T^{-1/2}$ comes entirely from the exact cancellation $\mathbb{E}[A_t m_{t+1} \mid \mathcal{F}_t] = 0$.

Term (B). $\|(\mathrm{B})\| \le T^{-1}\sum_{t\le T}\|I - P_{t-1}H\|_{\mathrm{op}}\|\delta_{t-1}\| = T^{-1}\sum_{t\le T} O(t^{-\nu_H})\|\delta_{t-1}\|$, which is a pathwise bound needing no lemma. With the rung-(iii) preliminary rate $\|\delta_t\| = O(t^{-1/2}\log^c t)$ this is $O(T^{-1} \cdot T^{1/2-\nu_H}\log^c T) = o(T^{-1/2})$ a.s.

Term (C). $r_t = (\widehat{H}_t - H)\delta_{t-1} + O(\|\delta_{t-1}\|^2)$, where $\widehat{H}_t$ is the block curvature. The quadratic part contributes $O(T^{-1}\sum_t \log t/t) = O(\log^2 T/T) = o(T^{-1/2})$. For the first part take $\zeta_t = \widehat{H}_t - H$, which is stationary, centred and square integrable under Assumption 3 (it is the block average of the terms appearing in (22) evaluated at $\theta^\star$, plus a Lipschitz remainder in θ_{t-1} bounded exactly as term (iv) there), and apply (16) with $\gamma_t \equiv 1$ and the $\mathcal{F}_{t-1}$-measurable weights acting by $A_t\zeta_t = P_{t-1}\zeta_t\delta_{t-1}$. By Lemma 2 and the rung-(iii) rate the envelope is $c_t = O(t^{q_{\mathrm{r}}} \cdot t^{-1/2}\log^c t)$ and the increments are $u_t = O(t^{-1}c_t)$, so

$$V_T = O\Big(\sum_{t\le T} t^{2q_{\mathrm{r}}-1}\log^{2c} t\Big) = O(T^{2q_{\mathrm{r}}}\log^{2c} T), \qquad U_T = O\Big(\sum_t t^{q_{\mathrm{r}}-3/2}\log^c t\Big) = O(1) \ \ (q_{\mathrm{r}} < 1/2),$$

and $\gamma_T c_T T^{1/2+\varpi} = O(T^{q_{\mathrm{r}}+\varpi}\log^c T)$. Hence the sum is $O(T^{q_{\mathrm{r}}+\varpi}\log^{c+1/2+\varpi} T)$ and (C) $= O(T^{q_{\mathrm{r}}-1+\varpi}\log^{c'} T) = o(T^{-1/2})$ a.s. because $q_{\mathrm{r}} < 1/2$. No independence between $\widehat{H}_t$ and δ_{t-1} is used or needed: adjacent blocks are only $O(1/b)$ apart in the sense of Lemma 12, not $O(\rho^b)$, and an earlier version of this proof wrongly appealed to the latter.

Substituting into (24) gives $\|\delta_T\|^2 = O(\log N_T/N_T)$ a.s. (Theorem 5) and $\sqrt{N_T}\delta_T = -N_T^{-1/2}H^{-1}\sum_{i\le N_T} g_i + o_p(1)$. The central limit theorem for strongly mixing stationary sequences with geometric coefficients and $2+\nu$ moments [39, 42] gives $N^{-1/2}\sum_{i\le N} g_i \rightsquigarrow \mathcal{N}(0, S)$, and Slutsky completes Theorem 6. The averaged variant follows from the same decomposition with $\gamma_t = ct^{-a}$ and the standard Abel argument for weighted averages [64, 71]. □

A.7 The L^2 rate (Theorem 9)

Take expectations in (24). The leading term has $\mathbb{E}\|N^{-1}H^{-1}\sum_{i\le N} g_i\|^2 = \mathrm{tr}(H^{-1}SH^{-1})/N + o(1/N)$ by Lemma 12, uniformly in b because it does not involve b at all. For the remainders, replace each almost sure bound above by its L^2 counterpart: the martingale parts are bounded in L^2 by their predictable variations, which were already computed in expectation in the proof of Lemma 14, and the coboundary parts by $\mathbb{E}\|z_1\|\sum_t u_t$. On the descent side, iterating (19) in expectation with $\mathbb{E}\|\bar{g}_t(\theta_{t-1})\|^2 \le C + C'\mathbb{E}[F(\theta_{t-1}) - F(\theta^\star)]$ (Assumption 2) and $\gamma_t^2\|P_{t-1}\|_{\mathrm{op}}^2 = O(t^{2q_{\mathrm{r}}-2})$ (Lemma 2) gives $\mathbb{E}[F(\theta_t) - F(\theta^\star)] = O(\log t/t)$ by the standard Chung-type recursion, and local strong convexity converts this to $\mathbb{E}\|\delta_t\|^2 = O(\log t/t) = O(\log N_t/N_t)\cdot b$; the extra factor b is absorbed because $N_t = bt$ and the leading term is b-free. All constants depend on b only through $\mathbb{E}\|\bar{g}_t\|^2 \le \mathbb{E}\|g_1\|^2$, which is monotone in b, so the bound is uniform over $b \le C\log N$. □

A.8 Coverage of dependence-blind intervals (Proposition 8)

By Theorem 6 and Theorem 3, $\sqrt{N}(\theta_{T,j} - \theta_j^\star) \rightsquigarrow \mathcal{N}(0, V_{jj})$ with $V = H^{-1}SH^{-1}$, while $[\bar{H}^{-1}\widehat{\Gamma}_0\bar{H}^{-1}]_{jj} \to_p [H^{-1}\Gamma_0 H^{-1}]_{jj} = V_{jj}/\kappa_j$. Hence the studentised statistic converges to $\mathcal{N}(0, \kappa_j)$ and $\mathbb{P}(|\mathcal{N}(0,\kappa_j)| \le z_{\alpha/2}) = 2\Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1$. For the homogeneous AR design, $\Gamma_k = \sigma_u^2(\phi\psi)^{|k|}\Sigma_x$ gives $S = \sigma_u^2\Sigma_x\sum_k(\phi\psi)^{|k|} = \sigma_u^2\Sigma_x(1+\phi\psi)/(1-\phi\psi)$ and $H = \Gamma_0/\sigma_u^2 = \Sigma_x$, so $V = \kappa\,\sigma_u^2\Sigma_x^{-1}$ with $\kappa = (1+\phi\psi)/(1-\phi\psi)$ and the inflation is the same in every coordinate. □

A.9 Long-run covariance estimation (Theorem 10)

(i) Bias of batch means. For the idealised estimator built from $\varepsilon_t = \bar{g}_t(\theta^\star)$, stationarity gives $\mathbb{E}[b\,\varepsilon_t\varepsilon_t^\top] = b\Lambda_0 = \sum_{|k|<b}(1-|k|/b)\Gamma_k$. Splitting,

$$b\Lambda_0 = S - \sum_{|k|\geq b}\Gamma_k - \frac{1}{b}\Big(\Delta - \sum_{|k|\geq b}|k|\Gamma_k\Big), \qquad \Delta = \sum_{k\geq 1}k(\Gamma_k + \Gamma_k^\top), \tag{25}$$

and by Lemma 12 both remainder sums are $O(b\rho^b)$, giving $\mathbb{E}[\widehat{S}^{\mathrm{BM}}] = S - \Delta/b + O(\rho^b)$ up to the drift term treated below.

(ii) Bias of the two-scale estimator. With $\Lambda_1 = \mathbb{E}[\varepsilon_0\varepsilon_1^\top] = b^{-2}\sum_{i\in B_0, j\in B_1}\Gamma_{j-i}$ and counting multiplicities of each lag (k for $1\leq k\leq b$, $2b-k$ for $b<k<2b$),

$$b(\Lambda_1 + \Lambda_1^\top) = \sum_{1\leq|k|\leq b}\frac{|k|}{b}\Gamma_k + \sum_{b<|k|<2b}\frac{2b-|k|}{b}\Gamma_k. \tag{26}$$

Adding (25) in the form $b\Lambda_0 = \sum_{|k|<b}(1-|k|/b)\Gamma_k$, the coefficient of Γ_k becomes exactly 1 for $|k|\leq b$ and $(2b-|k|)/b \in (0,1)$ for $b<|k|<2b$, so

$$b\Lambda_0 + b(\Lambda_1+\Lambda_1^\top) = S - \sum_{|k|\geq 2b}\Gamma_k - \sum_{b<|k|<2b}\Big(1 - \frac{2b-|k|}{b}\Big)\Gamma_k = S + O(\rho^b), \tag{27}$$

using Lemma 12. This is the flat-top window of Politis and Romano [73] at block resolution; the point is that under geometric mixing its bias is exponentially small in b, whereas the batch-means bias in (i) is only $O(1/b)$. The algebraic identity $\widehat{S}^{\mathrm{FT}} = 2\widehat{S}^{\mathrm{BM}}(2b) - \widehat{S}^{\mathrm{BM}}(b)$ (using disjoint pairs) is immediate from $\widehat{S}^{\mathrm{BM}}(2b) = \widehat{S}^{\mathrm{BM}}(b) + (b/T)\sum(\bar{g}_{2k-1}\bar{g}_{2k}^\top + \text{transpose})$.

(iii) Variance. $\widehat{S}^{\mathrm{BM}}_{jk}$ is an average of $T' \asymp N/b$ terms $q_t = b\,\bar{g}_{tj}\bar{g}_{tk}$. These are quadratic in the scores, so bounding their autocovariances by Lemma 12 needs $2+\delta$ moments of q_t, i.e. $4+2\delta$ moments of the scores; this is exactly why Assumption 5 asks for $\mathbb{E}\|g_1\|^{4+\nu'} < \infty$ with a margin $\nu' > 0$ rather than plain fourth moments. Given that, (q_t) is a measurable function of a geometrically mixing sequence at block resolution, hence itself geometrically mixing with $\alpha_{\mathrm{mix}}(k) \leq K\rho^{kb}$, and Davydov gives $|\operatorname{Cov}(q_t, q_{t+k})| \leq C\rho^{\delta kb/(2+\delta)}$. Summing the geometric series, $\operatorname{Var}[\widehat{S}^{\mathrm{BM}}_{jk}] = T'^{-2}\sum_{t,t'}\operatorname{Cov}(q_t, q_{t'}) = O(1/T') = O(b/N)$, and the same argument applies to the lag-one term $b\,T'^{-1}\sum\bar{g}_{2k-1,j}\bar{g}_{2k,k}$, hence to $\widehat{S}^{\mathrm{FT}}$.

Drift. Write $\bar{g}_t(\theta_{t-1}) = \varepsilon_t + \Xi_t$ with $\Xi_t = \widehat{H}_t\delta_{t-1} + O(\|\delta_{t-1}\|^2)$. Then $\widehat{S}^{\mathrm{FT}}$ built from $\bar{g}_t(\theta_{t-1})$ differs from the idealised version by terms bounded by $b\,T'^{-1}\sum_{t\in\mathcal{T}}\{\|\Xi_t\|^2 + 2\|\Xi_t\|\|\varepsilon_t\|\}$. Two facts are needed. First, on the retained window $t \asymp T$, Theorem 5 gives $\mathbb{E}\|\delta_{t-1}\|^2 = O(\log t/(bt))$. Second, $\widehat{H}_t$ is a b-sample estimate of a $d\times d$ matrix, so for δ independent of block t, $\mathbb{E}\|\widehat{H}_t\delta\|^2_{\mathrm{op}} = \delta^\top\mathbb{E}[\widehat{H}_t^2]\delta \leq \big(1 + Cd/b\big)\|H\delta\|^2_{\mathrm{op}}$ under Assumption 3 (the d/b term is the fluctuation $\mathbb{E}[(\widehat{H}_t - H)M(\widehat{H}_t - H)]$, whose trace contributes a factor $\operatorname{tr}(HM)/b$). Hence

$$b\,T'^{-1}\sum_{t\in\mathcal{T}}\mathbb{E}\|\Xi_t\|^2 = O\Big((1+d/b)\,T'^{-1}\sum_t \log t/t\Big) = O\big((1+d/b)\,b\log^2 N/N\big) = O\big((b+d)\log^2 N/N\big),$$

and the cross term is of smaller order by Cauchy–Schwarz. Writing $\tau_{b,N} = (b+d)\log^2 N/N$, this is dominated by the $\sqrt{b/N}$ standard deviation in (iii) whenever $b+d = o(\sqrt{N})$ up to logarithms; in the reported experiments $b+d \leq 340$ and $N \geq 2.5\times 10^4$. Note that $\tau_{b,N}$ grows as b falls below d, which is visible in Figure 2(b) and is why we do not recommend $b \ll d$.

(iv) Rate. The entrywise bounds give $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} \le \|\widehat{S}^{\mathrm{FT}} - S\|_F \le d \max_{jk} |\widehat{S}^{\mathrm{FT}}_{jk} - S_{jk}|$, hence $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} = O_p(d\sqrt{b/N} + \rho^b + \tau_{b,N})$; the factor d is the price of the conversion and we do not try to remove it (a matrix-Bernstein argument under an explicit sub-exponential condition would give $\sqrt{db/N}$). Choosing $b = \lceil c \log N \rceil$ with $c > 2/\log(1/\rho)$ makes $\rho^b = O(N^{-2})$ and gives $O_p(\sqrt{\log N/N})$. For $\widehat{S}^{\mathrm{BM}}$ the bias term $1/b$ replaces ρ^b and the trade-off $\sqrt{b/N} \asymp 1/b$ gives $b \asymp N^{1/3}$ and $O_p(N^{-1/3})$. Finally, $\bar{H}^{-1} \to_p H^{-1}$ (Theorem 3) and $\widehat{S}^{\mathrm{FT}} \to_p S$ give $u_j^\top \widehat{S}^{\mathrm{FT}} u_j \to_p V_{jj}$, and Theorem 6 with Slutsky gives the studentised limit. □

A.10 Curvature schedules (Proposition 11)

Let $\widehat{H}^{\mathrm{gap}} = n^{-1} \sum_{j \le n} Y_{jm}$. By stationarity,

$$n \operatorname{Var}[\widehat{H}^{\mathrm{gap}}] = \Gamma_0^H + 2 \sum_{j \ge 1} \Big(1 - \frac{j}{n}\Big) \Gamma_{jm}^H = \Gamma_0^H + 2 \sum_{j \ge 1} \Gamma_{jm}^H + o(1) = \Gamma_0^H + O(\rho^m),$$

using $\|\Gamma_k^H\|_{\mathrm{op}} \le C\rho^k$ (Lemma 12 applied to Y, which has $\eta' > 2$ moments by Assumption 3). For Bernoulli thinning, let $Z_i \overset{iid}{\sim} \mathrm{Bernoulli}(p)$ be independent of the data and let

$$\widehat{H}^{\mathrm{Bern}} = \Big(\sum_{i \le N} Z_i\Big)^{-1} \sum_{i \le N} Z_i Y_i$$

be the estimator with the *realised* normaliser — which is what the algorithm computes, since W counts the updates actually made. Using a fixed normaliser pN instead would leave a term $p(1-p)\,\mathrm{vec}(H)\mathrm{vec}(H)^\top$ coming from $\mathbb{E}[Y_i] = H \ne 0$, which is $\Theta(1)$ and would swamp the $O(1/m)$ effect; the ratio form removes it, because $\widehat{H}^{\mathrm{Bern}} - H = (\sum_i Z_i)^{-1} \sum_i Z_i (Y_i - H)$ *exactly*. Writing $n_Z = \sum_i Z_i$ with $\mathbb{E} n_Z = pN = n$ and $n_Z/n \to 1$ a.s., the delta method for a ratio gives $n \operatorname{Var}[\widehat{H}^{\mathrm{Bern}}] = n^{-1} \operatorname{Var}\big[\sum_i Z_i(Y_i - H)\big] + o(1)$, and

$$\operatorname{Var}\Big[\sum_i Z_i (Y_i - H)\Big] = \sum_i p\,\Gamma_0^H + \sum_{i \ne j} p^2 \Gamma_{i-j}^H = Np\,\Gamma_0^H + 2p^2 \sum_{k \ge 1} (N-k) \Gamma_k^H.$$

Dividing by $n = pN$ and using $p = 1/m$, $n \operatorname{Var}[\widehat{H}^{\mathrm{Bern}}] = \Gamma_0^H + 2p \sum_{k \ge 1} \Gamma_k^H + o(1) = \Gamma_0^H + m^{-1}(S^H - \Gamma_0^H) + o(1)$. □

Remark 7. The contrast is that random thinning leaves a fixed fraction p of *all* lag pairs in the sum, whereas a deterministic gap removes every lag below m. For Gaussian AR(1) covariates and $\alpha \equiv 1$, Γ_k^H decays like ϕ^{2k} and $\kappa_H = (1+\phi^2)/(1-\phi^2)$.

B Experimental details

B.1 Data-generating processes and their population targets

Every simulated design pairs a dependent covariate process with a dependent error or latent process, for the reason given in Remark 1. In all four, $\theta^\star = (-1, \dots, 1)$ evenly spaced in $\mathbb{R}^d$, $d = 20$, and the covariate covariance is the ill-conditioned $\Sigma = Q \operatorname{diag}((i/d)^c) Q^\top$ of Boyer and Godichon-Baggioni [9] with Q a random orthogonal matrix, $c = 2$ (condition number 400) except for the logistic design where $c = 1$ (condition number 20) to keep the linear predictor in a sensible range. The population (H, Γ_0, S) is fixed once per design; Monte Carlo replicates vary only the realised path, which is checked in `tests/test_oracles.py`.

AR-hom (closed form). $x_t = \phi x_{t-1} + \sqrt{1-\phi^2}\Sigma^{1/2}\eta_t$, $u_t = \psi u_{t-1} + \sqrt{1-\psi^2}\sigma_u\varepsilon_t$, $y_t = x_t^\top\theta^\star + u_t$, with η, ε standard normal and independent, $\phi = \psi = 0.7$, $\sigma_u = 1$. Then $H = \Sigma$, $\Gamma_k = \sigma_u^2(\phi\psi)^{|k|}\Sigma$ and $S = \sigma_u^2\Sigma(1+\phi\psi)/(1-\phi\psi)$, so $\kappa_j \equiv 2.922$.

AR-het (closed form). $x_t = A_\phi x_{t-1} + \Sigma_\eta^{1/2}\eta_t$ with $A_\phi = \mathrm{diag}(\phi_1, \ldots, \phi_d)$, ϕ_j evenly spaced on $[0, 0.9]$. The stationary covariance is the Hadamard product $\Sigma_x = \Sigma_\eta \circ [1/(1-\phi_i\phi_j)]$, which is positive definite because $[1/(1-\phi_i\phi_j)] = \sum_{k\geq 0}(\phi\phi^\top)^{\circ k}$ is a sum of Hadamard powers of a rank-one positive semidefinite matrix whose first d terms are linearly independent when the ϕ_j are distinct. With AR(1) errors, $\mathbb{E}[x_t x_{t+k}^\top] = \Sigma_x A_\phi^k$, $\Gamma_k = \sigma_u^2\psi^k\Sigma_x A_\phi^k$ and $S = \sigma_u^2(\Sigma_x + \Sigma_x G_\psi + G_\psi\Sigma_x)$ with $G_\psi = \psi A_\phi(I - \psi A_\phi)^{-1}$. Because the ϕ_j differ, κ_j varies across coordinates, so a scalar correction cannot pass this design.

Markov (closed form). s_t is a $K = d+1$ state Markov chain with $\mathbb{P}(s_t = s_{t-1}) = 0.97$, uniform off-diagonal transitions and uniform stationary law; $x_t = V_{\cdot,s_t} + \Sigma_\nu^{1/2}\nu_t$, so covariates are a Markov-modulated Gaussian mixture—non-Gaussian marginals, discrete latent state, regime (non-linear) dependence. The errors are AR(1) with $\psi = 0.8$. Then $\Sigma_x = V\mathrm{diag}(\pi)V^\top + \Sigma_\nu$, $\mathbb{E}[x_t x_{t+k}^\top] = V\mathrm{diag}(\pi)P^k V^\top$ and $S = \sigma_u^2(\Sigma_x + R + R^\top)$ with $R = V\mathrm{diag}(\pi)\{\psi P(I-\psi P)^{-1}\}V^\top$.

The choice of centres matters and is worth recording, because the obvious construction produces a design that *looks* strongly dependent but has no inferential consequence. With randomly drawn centres and $K \ll d$, the regime component $V\mathrm{diag}(\pi)V^\top$ has rank K and sits in high-curvature directions of Σ_x; the sandwich $H^{-1}\Gamma_k H^{-1}$ is then negligible against $H^{-1}\Gamma_0 H^{-1}$, whose mass is in the *low*-curvature directions, and κ_j came out in $[1.00, 1.05]$ for every setting we tried. We therefore take $K = d+1$ and the centres to be a regular simplex scaled by the target covariance: $V = \sqrt{K}\,\mathrm{chol}(\Sigma)U$ with $U \in \mathbb{R}^{d\times K}$ an orthonormal basis of $\mathbf{1}^\perp$ in $\mathbb{R}^K$, which gives $V\pi = 0$ and $V\mathrm{diag}(\pi)V^\top = \Sigma$ exactly. The regime component is then full rank and proportional to Σ, so $R_k = \lambda_2^k\Sigma$ with λ_2 the chain's second eigenvalue and $\kappa_j \to (1+\psi\lambda_2)/(1-\psi\lambda_2)$ as $\Sigma_\nu \to 0$: at our settings, $\kappa_j = 7.55$ in every coordinate against the noise-free limit 7.88.

This design is the one that forces both the longer warm-up and the stability shift, and the same feature causes both. With $\mathbb{P}(s_t = s_{t-1}) = 0.97$ the chain dwells in a regime for about 33 observations and its integrated autocorrelation time is $(1+0.97)/(1-0.97) \approx 66$, so a block of $b = 20$ consecutive observations almost always lies inside a single regime and the block Hessian $\widehat{H}_t$ is nearly rank one. Two consequences, which we separated only after mistaking the second for the first. A warm-up of $50d$ *consecutive* observations spans only about thirty regime visits, so it leaves $\bar{H}$ rank-starved; $200d$ fixes that (Table 5(c′)). But a well-estimated $\bar{H}$ is not enough, because the near-rank-one $\widehat{H}_t$ makes $\|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}}$ large — median 33 here, against 20 on AR-hom, with block maxima 89 and 55 — so the contraction condition (8) fails at $\gamma_1 = 1$ and the first few steps overshoot by orders of magnitude. At $c_{\mathrm{w}} = 200$ and no shift, the relative error of the point estimate still exceeds one on 7 of 10 replicates. Both fixes are needed, and neither substitutes for the other (Table 5(i)).

Logistic (simulated oracle). x_t is AR(1) with $\phi = 0.7$; the latent error is $e_t = \mathrm{logit}(\Phi(\nu_t))$ with ν_t a unit-variance AR(1) with $\psi = 0.8$, so e_t has an exact standard logistic marginal and $y_t = \mathbf{1}\{x_t^\top\theta^\star + e_t > 0\}$ satisfies $\mathbb{P}(y_t = 1 \mid x_t) = \sigma(x_t^\top\theta^\star)$ *exactly*. The working logistic likelihood is therefore correctly specified marginally—$\theta^\star$ is the true parameter and $\Gamma_0 = H$ by the information equality—while the score is serially correlated through ν. This is the marginal-model situation of Liang and Zeger [59]. H is computed to machine precision by Gauss–Hermite quadrature using the decomposition $x = (\Sigma_x\theta^\star/\sigma_\ell^2)\ell + x_\perp$ with $\ell = x^\top\theta^\star$ the linear predictor and $x_\perp \perp \ell$; S has no closed form and is obtained from eight independent streams of 10^6 observations with the score evaluated at the known $\theta^\star$ and a Bartlett long-run covariance at lag 100 (the score autocovariance decays like $\psi^k = 0.8^k$, so the truncation error is below 10^{-9}). The resulting relative Monte Carlo error of the sandwich diagonal is 7.5×10^{-3}; it is the only non-exact population quantity in the paper.

B.2 Real streams: features, segments and Protocol B's nominal level

Responses. Both responses are log-transformed and then standardised to zero mean and unit variance, exactly as the covariates are. The standardisation is not cosmetic. Without it the mean of log(traffic volume), about 7.9, sits entirely in the intercept, which is then roughly 400 standard errors from the $\theta_0 = 0$ that a streaming method starts from; a $1/t$ schedule needs of order that many steps merely to cover the distance, so on a 4000-hour segment (which affords only about 100 blocked steps at $b = 40$) the experiment would be measuring the distance to the initialisation rather than anything about inference — the infeasible oracle-variance interval covered 0.14 in that configuration, which is the diagnostic that the point estimate, not the covariance, was at fault. Standardising the response is an affine reparametrisation of a linear model and changes no coverage statement; it is also what any practitioner would do.

Features. Traffic: intercept, standardised temperature and cloud cover, two daily harmonics and one weekly harmonic, a holiday indicator propagated from the labelled midnight row to the whole calendar day, and a weekend indicator ($d = 11$). Rainfall is *excluded*: `rain_1h` is identically zero over stretches of several thousand consecutive hours, so $\log(1 + \text{rain})$ is a constant column on some contiguous segments, which makes the design there exactly rank deficient and the per-segment population targets undefined. Air quality: intercept, standardised temperature, pressure, dew point, $\log(1 + \text{rain})$ and wind speed, plus the same harmonics ($d = 12$). Feature construction was fixed before any response was looked at, and `exp6_real.py` asserts that every evaluation segment has full rank, reporting the worst per-segment condition number.

Segments and hold-out. The leading 15% of each stream is reserved for choosing the first-order baselines' step-size constants and is excluded from every evaluation segment; the constants are chosen per stream, and the grid and whether the optimum is interior are recorded in `results/exp6_real.json`. Evaluation segments are the disjoint contiguous blocks that follow.

Protocol B's nominal level. Let R be the number of evaluation segments, all of the same length, let $\hat{\theta}_k$ be the estimate from segment k and $\hat{\theta}_{\text{full}}$ the full-stream M-estimator. If (i) the segment estimates are approximately uncorrelated, (ii) they have a common variance V, and (iii) $\hat{\theta}_{\text{full}} \approx R^{-1} \sum_k \hat{\theta}_k$, then

$$\operatorname{Var}(\hat{\theta}_k - \hat{\theta}_{\text{full}}) = \operatorname{Var}(\hat{\theta}_k) - \operatorname{Var}(\hat{\theta}_{\text{full}}) = V(1 - 1/R),$$

so an interval built to cover $\theta^\star$ with probability $1 - \alpha$ covers $\hat{\theta}_{\text{full}}$ with probability $2\Phi\big(z_{\alpha/2}/\sqrt{1 - 1/R}\big) - 1$, which is what the header of Table 4 reports. Assumption (iii) requires the full-stream estimator to be (approximately) the equally weighted average of the segment estimators, which holds when the segments have equal length and similar $\widehat{H}$; (i) is the same approximation that makes segments usable as replicates at all. The correction is small but not negligible—at the R of our traffic stream it raises the target from 0.950 to 0.976, and at the R of the air-quality stream to 0.994—and both the corrected target and the raw coverage appear in Table 4, computed from R rather than quoted from memory. The correction is also least trustworthy exactly where it is largest: with only a handful of segments per stream, assumption (iii) is a crude approximation and the corrected target is close to one, which compresses the comparison. That is the main reason we treat Protocol A, whose target is exact by construction, as the primary real-data evidence and Protocol B as the check that the conclusion survives a real error process.

B.3 Hyperparameters

Unless a sweep says otherwise: block length $b = 20$ $(= d)$, gap $m = b$ (one curvature slot per block), no Bernoulli thinning $(p = 1)$, log-weights off $(w' = 0)$, ridge $c_\iota = 0.01$, $q_\mathrm{r} = 0.4$, initial curvature scale $\lambda_0 = 10^{-6}$ (relative to the first block's mean curvature magnitude, so it is dimensionless), warm-up $c_\mathrm{w} = 200$ observations per parameter capped at $\varpi_\mathrm{w} = 10\%$ of the stream, retained window dyadic (last half to last three quarters of blocks), step-length safeguard constant $c_D = 1$ in (6), and the schedule shift of Appendix C.1 switched off (`t0_mult` = 0). Table 5 varies each of these. The warm-up constant is 200 rather than the 50 that suffices on the autoregressive designs because of the regime-switching design, for the reason given in Appendix B.1. The warm-up and the safeguard address different failures and neither substitutes for the other: the warm-up makes $\bar{H}$ informative, and the safeguard makes the first steps safe to take even when it is. With $c_\mathrm{w} = 200$ and no safeguard the regime-switching design still diverges (Table 5(i)); with the safeguard and $c_\mathrm{w} = 50$ it no longer diverges but the curvature is still rank-starved, and the relative variance is 2.19 against 1.41 at $c_\mathrm{w} = 200$ (Table 5(c′)). The two devices are complementary in exactly that way: the safeguard turns a divergence into an inefficiency, and the warm-up removes the inefficiency. The 10% cap on the warm-up never binds in the simulations, where $N = 200{,}000$ and $c_\mathrm{w} d \le 4000$; it binds on every real-data segment, where N is 8,000 or 10,000 (Table 5(j)).

The dependence-blind competitor "streaming Newton, i.i.d. theory" uses the base method's schedule exactly: every observation is a candidate curvature slot $(m = 1)$ with Bernoulli retention $p = 1/d$, and the interval uses the i.i.d. plug-in $\widehat{\Gamma}_0 = N^{-1} \sum_i s_i(\theta_{t-1}) s_i(\theta_{t-1})^\top$, accumulated over exactly the observations the blocked estimators use—that is, excluding the warm-up, so that the comparison is not contaminated by scores evaluated at θ_0.

First-order baselines use step $\gamma_t = c\,t^{-3/4}$ (averaged SGD) or a constant c with AdaGrad scaling, with c chosen from a 19-point logarithmic grid on $[10^{-3}, 10^{1.5}]$ minimising squared error on eight *pilot* streams whose seeds are disjoint from the evaluation streams; the selected constants are recorded in `results/exp1_main.json`. They are additionally handed the *true* Hessian for their sandwich. Random scaling uses the two-sided 95% quantile 6.811 of the pivot $|W(1)|/(\int_0^1 (W(r) - rW(1))^2 dr)^{1/2}$, taken from Kiefer and Vogelsang [47] as tabulated by Lee et al. [54, Table 2]. Our own simulation of that quantile, using the exact Karhunen–Loève representation $\int_0^1 B^2 = \sum_{k\ge1} Z_k^2/(k\pi)^2$ with Z_k i.i.d. standard normal independent of $W(1)$, gives 6.746 from 4×10^6 draws (the sampler is validated by reproducing $\mathbb{E}\int_0^1 B^2 = 1/6$ and $\mathrm{Var}\int_0^1 B^2 = 1/45$); we use the larger published value, which is generous to that baseline. We also flag a normalisation trap: several papers quote 5.32 as "the 95% critical value", which is the 95% quantile of the *signed* statistic, i.e. the two-sided 90% value, and using it would make the random-scaling intervals substantially too narrow.

The iterate-path comparator. Figure 3(a) compares against overlapping batch means computed from the iterate path, in the matrix form of the estimator used in the dependent-data literature: with T recorded iterates, window length b_n, window averages $\bar{\theta}_{b_n,t} = b_n^{-1} \sum_{k=t}^{t+b_n-1} \theta_k$ and overall average $\bar{\theta}$,

$$\widehat{\Sigma}^{\mathrm{it}} = \frac{N}{T} \cdot \frac{b_n}{T - b_n + 1} \sum_t \big(\bar{\theta}_{b_n,t} - \bar{\theta}\big)\big(\bar{\theta}_{b_n,t} - \bar{\theta}\big)^\top, \qquad b_n = \lceil T^{3/4} \rceil,$$

which estimates the covariance of $\sqrt{N}(\bar{\theta} - \theta^\star)$, i.e. the same Σ our sandwich estimates. It uses no gradients and no curvature estimate. We compute it from our own recorded trajectory, with cumulative sums, so it costs $O(Td)$ time and $O(Td)$ memory; it is a reference, not a streaming method, and we make no claim about the constants of any particular published recursion.

B.4 Reproduction and compute

Everything in this paper is public: the estimator, the experiment scripts, the saved result files from which every number, figure and table is generated, and the tests that validate every closed-form population quantity against long simulation. The exact release used to produce the results reported here is archived with a persistent identifier [46],

`10.5281/zenodo.21891313` (development version:
`https://github.com/Kendiukhov/streaming-quasi-newton-dependent-data`),

and the concept identifier `10.5281/zenodo.21891312` always resolves to the most recent release. The two real data sets are third-party and are not redistributed; `fetch_data.py` downloads them from the UCI Machine Learning Repository [14, 40, 45]. `make all` runs everything. All BLAS libraries are pinned to a single thread and parallelism is over Monte Carlo replicates (`experiments/_env.py` explains why: on the machine used, an unpinned tall-skinny $(d, N) \times (N, d)$ product with $N \approx 4 \times 10^4$, $d \approx 12$ took 18 s against 5 ms pinned). Total compute for every number in the paper is a few CPU-hours. The machine was shared and its load varied by more than an order of magnitude during the runs, so Table 6 reports (i) exact machine-independent operation counts computed from the algorithm itself and (ii) wall-clock times measured with the repetitions *interleaved* across methods and summarised by the minimum over repetitions—round-robin ordering exposes every method to the same load pattern, and external load can only make a measurement slower, so the minimum is the robust summary. The operation counts are the authoritative statement; the times corroborate them.

B.5 Additional results

C Safeguarding the blocked step

Proof of Proposition 1. On $\mathcal{A}$ there are finitely many t at which Π_t is active, so $T_0 = \max\{t : \Pi_t \text{ active}\}$ is finite and for $t > T_0$ the recursion is $\theta_t = \theta_{t-1} - \gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t(\theta_{t-1})$ verbatim. Two sequences that agree from a finite index on have the same limit points, the same almost sure rate and the same limit law, since all three are tail properties; the initial condition at T_0 is a finite random vector, and Theorems 4 to 6 hold for the recursion started from any initial condition (Assumption 4 is global). For the last claim, if $\theta_t \to \theta^\star$ and $\|\gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t(\theta_{t-1})\| \to 0$ almost surely then for almost every ω there is $T_1(\omega)$ beyond which $\|\gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t\| < c_D(1 + \|\theta^\star\|)/2 \le c_D(1 + \|\theta_{t-1}\|)$, so Π_t is inactive for $t > T_1$ and $\omega \in \mathcal{A}$. □

We repeat here what Section 2 says, because it is the kind of gap a reader is entitled to be told about twice. Proposition 1 is conditional on $\mathcal{A}$ and does not establish $\mathbb{P}(\mathcal{A}) = 1$ unconditionally: the implication runs from convergence to eventual inactivity, not the other way. The reason we do not close the loop is that the safeguard factor $\min\{1, c_D(1 + \|\theta_{t-1}\|)/\|v\|\}$ depends on $\bar{g}_t$, hence is $\mathcal{F}_t$- and not $\mathcal{F}_{t-1}$-measurable, and multiplying the noise term $\gamma_t \bar{H}^{-1} \varepsilon_t$ by an $\mathcal{F}_t$-measurable factor destroys the martingale-difference property that Theorems 4 and 6 rest on. Two routes would close it and both are outside our scope: the expanding-truncation construction of Kushner and Yin [52] and Andrieu et al. [4], which replaces the cap by a projection onto a ball of radius increasing with the number of activations and proves the activation count finite; or an $\mathcal{F}_{t-1}$-measurable variant that caps using the *previous* block's amplification, $\gamma_t \wedge \tau / \|\bar{H}_{t-2}^{-1} \widehat{H}_{t-1}\|_{\mathrm{op}}$, which is predictable and so leaves the martingale structure intact, at the cost of an extra $O(d^2)$ per block. What we do instead is report the activation counts, which are between 2 and 7 per run and do not grow with N.

C.1 The step-size shift: the alternative we implemented and rejected

The variant replaces the step sequence $\gamma_t = c\, t^{-a}$ of (6) by $\gamma_t = c\,(t + t_0)^{-a}$ with

$$t_0 = \max\left\{0,\ \tfrac{c}{2} \max_{t \le T_{\mathrm{w}}} \|\bar{H}_{T_{\mathrm{w}}}^{-1} \widehat{H}_t\|_{\mathrm{op}} - 1\right\}, \tag{28}$$

$T_{\mathrm{w}} = \lceil c_{\mathrm{w}} d/b \rceil$ being the last warm-up block, and switches the step-length safeguard Π_t off. By construction (8) then holds at $t = 1$ for every block whose amplification does not exceed the warm-up maximum. The spectral norms are obtained by four power iterations per warm-up block with $\widehat{H}_t$ applied in factored form, so no $d \times d$ product is formed and the whole computation costs $O(c_{\mathrm{w}}(d^2 + db)\, d/b)$ flops once, $O(1)$ in N; with $b = m = d$ that is $O(c_{\mathrm{w}} d^2)$, some 4–5% of the total at the settings of Table 6. Setting `t0_mult` = 0.5 and `step_cap` = 0 in the code selects this variant.

Correctness of the rejected shift variant. Write T_{w} for the last warm-up block. For every $t \le T_{\mathrm{w}}$, $\|\bar{H}_{T_{\mathrm{w}}}^{-1} \widehat{H}_t\|_{\mathrm{op}} \le \|\bar{H}_{T_{\mathrm{w}}}^{-1}\|_{\mathrm{op}} \|\widehat{H}_t\|_{\mathrm{op}}$. The first factor is finite almost surely by Lemma 2, which bounds it by $c_1^{-1} n_{T_{\mathrm{w}}}^{q_{\mathrm{r}}}$ using only the ridge in (4) and no property of the iterates. The second is finite almost surely because $\|\widehat{H}_t\|_{\mathrm{op}} \le b^{-1} \sum_{i \in B_t} \alpha_i \|\Psi_i\|^2$ and Assumption 3 gives $\mathbb{E}\|\alpha \Psi \Psi^\top\|_{\mathrm{op}}^{\eta'} \le C_{\eta'}$ with $\eta' > 2$, so each summand is finite a.s. and the maximum is over the finitely many blocks $t \le T_{\mathrm{w}}$. Hence $t_0 < \infty$ a.s. Measurability with respect to $\mathcal{F}_{T_{\mathrm{w}}} = \sigma(\xi_1, \ldots, \xi_{bT_{\mathrm{w}}})$ is immediate: (28) is a function of $\bar{H}_{T_{\mathrm{w}}}$ and of $\{\widehat{H}_t\}_{t \le T_{\mathrm{w}}}$, all of which are built from the warm-up window and from θ_0, and the warm-up takes no parameter step.

For the second claim, fix ω in the almost sure event $\{t_0 < \infty\}$ and let $K = \lceil t_0 \rceil$. Then $c(t + t_0)^{-a} \ge c(t + K)^{-a}$, and $\sum_{t \ge 1}(t + K)^{-a} = \sum_{t > K} t^{-a} = \infty$ for $a \le 1$, so $\sum_t \gamma_t = \infty$. Likewise $c(t + t_0)^{-a} \le ct^{-a}$, so $\sum_t \gamma_t^{1+\delta} \le c^{1+\delta} \sum_t t^{-a(1+\delta)} < \infty$ whenever $a(1 + \delta) > 1$, which is the same condition as for the unshifted sequence. Finally $(t + t_0)^{-a}/t^{-a} = (1 + t_0/t)^{-a} \to 1$. Conditioning on $\mathcal{F}_{T_{\mathrm{w}}}$ is harmless because $\mathcal{F}_{T_{\mathrm{w}}}$ is generated by finitely many observations, so it changes no tail statement and, by Assumption 5, leaves the stream after the warm-up geometrically mixing with the same constants up to a factor.

Theorem 4 and the preliminary rate use the step sequence only through $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 < \infty$, both just verified, so they transfer immediately. Theorems 5 and 6 and Proposition 7 deserve one extra line, because their proof uses the identity $\gamma_t = 1/t$ explicitly in the Abel summation (23), and we would rather write the shifted identity out than assert that nothing changes. With $\gamma_t = (t + t_0)^{-1}$, multiplying $\delta_t = \delta_{t-1} - \gamma_t P_{t-1}\{H\delta_{t-1} + \varepsilon_t + r_t\}$ by $t + t_0$ gives

$$(t + t_0)\, \delta_t = (t - 1 + t_0)\delta_{t-1} + (I - P_{t-1}H)\delta_{t-1} - P_{t-1}\varepsilon_t - P_{t-1} r_t, \tag{29}$$

which is (23) with t replaced by $t + t_0$ throughout. Summing from $t = 1$ to T now telescopes to $(T + t_0)\delta_T = t_0 \delta_0 + \sum_{t \le T}\{\cdot\}$ instead of $T\delta_T = \sum_{t \le T}\{\cdot\}$, so dividing by $T + t_0$,

$$\delta_T = \frac{t_0\, \delta_0}{T + t_0} - \frac{1}{T + t_0} H^{-1} \sum_{t \le T} \varepsilon_t + \frac{T}{T + t_0}\big\{(\mathrm{A}) + (\mathrm{B}) - (\mathrm{C})\big\},$$

with (A), (B), (C) exactly as in the unshifted proof. Two differences from (23), and both are negligible. The new first term is $O(t_0/T) = O(1/T) = o(T^{-1/2})$ because t_0 is a finite constant, so it does not enter the limit. And $T/(T + t_0) = 1 + O(1/T)$ and $(T + t_0)^{-1} = T^{-1}(1 + O(1/T))$, so every term of the unshifted decomposition is multiplied by $1 + O(1/T)$; since each of them is $O_p(T^{-1/2})$ or smaller, the perturbation is $O_p(T^{-3/2})$. The remaining estimates — the bounds on (A), (B) and (C), the martingale central limit theorem applied to $T^{-1/2} \sum_{t \le T} \varepsilon_t$, and the Gordin decomposition — do not involve γ_t at all. Hence the rate, the block-freeness statement and the limit law are unchanged. So the shift variant has the cleaner theory of the two; it is the empirical comparison, not the theory, that made us prefer the safeguard. □

Two remarks on what this variant does and does not do. First, (28) does not make (8) hold for *every* t: the maximum in (28) is over warm-up blocks, so a later block with an unusually badly conditioned $\widehat{H}_t$ can still violate (8) at that single t. What the shift buys is that the violation happens with $\gamma_t \leq (1 + t_0)^{-1}$ rather than $\gamma_1 = 1$, so a single bad block moves θ by a bounded multiple of its distance to $\theta^\star$ instead of by a large one, and the recursion recovers. Second, the proposition is about asymptotics; the finite-sample effect of the shift is a trade, because a larger t_0 also slows the decay of the initial condition, which decays like $t_0/(T + t_0)$ rather than $1/T$. This is why we set t_0 from (8) — the smallest value that restores contraction — rather than by taking a comfortable margin, and why Table 5 reports the sensitivity.